

Nationalistic Labor Policies Hinder Financial Innovation*

Francesco D’Acunto[†], Hengyi Huang[‡], Michael Weber[§], Jin Xie[¶], Liu Yang^{||}

November 16, 2025

Abstract

Hiring restrictions on high-skilled foreign workers hinder financial innovation without increasing domestic high-skill employment. Using variation from the Employ American Workers Act (EAWA), which prohibited TARP banks from hiring new foreign workers until TARP repayment, we show that banks relying on foreign inventors pre-crisis experienced greater declines in patent quantity and quality, particularly in FinTech, cybersecurity, and payments. First-time foreign inventor patents dropped the most. Banks neither replaced internal innovation with external patents nor substituted foreign talent with domestic hires. Instead, affected banks retained pre-crisis foreign workers by offering higher wage premia over average domestic wages.

JEL Classification: G28, J21, K23, O31.

Keywords: FinTech, Cybersecurity, Labor & Finance, Human Capital, Financial Regulation, IO & Finance, Patents, Immigration, TARP.

*For very helpful comments, we thank Shai Bernstein, Allen Berger, Fabio Braggion, Ben Charoenwong, Jess Cornaggia, Jesus Gorrin, Julio Crego, Stephen Dimmock, Joost Driessen, Michael Faulkender, Haoyu Gao, Umit Gurun, Iftekhar Hasan, Sabrina Howell, Gerard Hoberg, Mark Jansen, Gazi Kabas, Ankit Kalda, Jens Kvaerner, Sung Kwan Lee, Robert Marquez, Ronald Masulis, Adair Morse, Vincenzo Pezone, Nicola Pavanini, Haluk Unal, Raluca Roman, Amirabas Salarkia, Johan Sulaeman, Sheridan Titman, Cara Vansteenkiste, Felipe Varas, Ren Xiao, Alminas Žaldokas, Hong Zhang, Xiaodong Zhu, and seminar participants at the Tilburg University, the University of Sydney Business Financing and Banking Research Workshop, ABFER 10th Annual Conference, the 2021 American Finance Association Annual Meeting, the 2023 China International Conference in Finance, the 3rd CUHKSZ-PHBS Economics and Finance Workshop, the European Finance Association 51st Annual Meeting, the 22nd FDIC Bank Annual Research Conference, the 2024 Finance Down Under Conference, the 8th Indonesian Financial Management Association (IFMA) International Conference, the 2nd International Economics Joint Conference “Navigating Turbulence in the Global Economy: Challenges and Opportunities”, and the UT Dallas 2024 Fall Finance Conference. Yuan He, Yuhuan He, and Yongxue Lu provided excellent research assistance. D’Acunto, Weber, and Xie thank the General Research Fund of the Hong Kong Research Grants Council (Project Number: 14507417) for financial support.

[†]McDonough School of Business, Georgetown University. e-Mail: francesco.dacunto@georgetown.edu

[‡]Department of Finance, Tilburg University. e-Mail: h.huang_1@tilburguniversity.edu

[§]Purdue University Daniels School of Business and NBER. e-Mail: weber366@purdue.edu

[¶]Peking University HSBC Business School, jinxie@phbs.pku.edu.cn

^{||}R.H.Smith School of Business, University of Maryland. e-Mail: lyang1@umd.edu

Hiring American workers for limited available jobs should be a top priority for businesses taking taxpayer money through the TARP bailout program. [...] there is no need for companies to hire foreign guest workers through the H-1B program when there are plenty of qualified Americans looking for jobs.
Senator C. Grassley (February 6, 2009)

Buried in the American Recovery and Reinvestment Act of 2009 was the Employ American Workers Act... This act foolishly hurt hundreds of finance companies by limiting their talent pool precisely when they needed new talent the most.

Wall Street Journal (June 20, 2012)

1 Introduction

Innovation is a fundamental driver of economic growth (e.g., [Romer, 1990](#); [Aghion and Howitt, 1992](#); [Jones, 1995](#); [Griliches, 2007](#); [He and Tian, 2020](#)) and is increasingly important to preserve national security in the information economy (e.g., [Weiss, 2014](#)). Research shows that high-skilled immigrants play an important role in fostering a country’s innovation by contributing talent and specialized knowledge that is not always available domestically.¹ At the same time, over the last decade, restrictions on both high- and low-skilled immigration have been at the forefront of the global political agenda. Support for these restrictions is often based on the premise that the reservation wage of foreign workers is lower than that of similarly skilled domestic workers, and hence, allowing the immigration of foreign workers forces domestic ones to either accept lower salaries or face unemployment.² These opposing views stem from different assumptions regarding the substitutability of skills between immigrant and domestic workers. High-level skills, such as those needed to produce innovation, are typically costly to build, difficult to replicate, and scarce in any individual domestic labor market.

To empirically assess the effects of foreign high-skill hiring restrictions, an ideal experiment would randomly ban a group of domestic firms from hiring foreign high-skilled workers and compare the innovation output and wage offers of such banned firms with those of other firms. If specialized skills are available domestically, the innovation of banned firms would not decline as long as the value of domestic wages does not exceed the expected value of the innovation that accrues to firms. In this case, firms would simply face higher wage costs to replace foreign skills with domestic skills. In contrast, if specialized skills

¹For instance, see [Hunt and Gauthier-Loiselle \(2010\)](#), [Hunt \(2011\)](#), [Borjas and Doran \(2012\)](#), [Moser et al. \(2014\)](#), [Laeven et al. \(2015\)](#), [Akcigit et al. \(2018\)](#), [Bernstein et al. \(2019\)](#), and [D’Acunto and Frésard \(2019\)](#).

²Using large-scale surveys and experiments across six countries, [Alesina et al. \(2023\)](#) discover prevalent misperceptions about the skill levels of immigrants relative to domestic workers and examine how these misperceptions influence domestic citizens’ support for various redistribution policies. [Cornaggia et al. \(2024\)](#) offer a comprehensive analysis of how unauthorized immigration impacts local government finances via municipal bond markets. They show the economic effects vary based on local labor-market conditions and institutional factors.

are unavailable domestically, the innovation of banned firms might decline, and they would still have to pay higher wages to retain the foreign talent that was employed before the ban, as these workers gain greater bargaining power after the ban (Glennon, 2024b).

In this paper, we consider a natural experiment that shares some features with the ideal experiment—a US regulation that banned some domestic banks, but not others, from hiring new foreign workers over staggered time periods—to test the effects, if any, of hindering firms from hiring new high-skilled immigrants on innovation quantity and quality and high-skill wages. Specifically, we exploit the *Employ American Workers Act* (EAWA), which in 2009 suddenly banned new foreign hires by US financial institutions that had previously enrolled in the *Troubled Asset Relief Program* (TARP) after the fall of Lehman Brothers without knowing that such a ban would have been imposed on them at a later date.³ We find that after the EAWA ban, relative to unaffected banks, banned banks reduced their innovation in terms of both quantity (number of patents filed) and quality (filed patents’ citations). Moreover, they began offering higher wage premia to retain foreign inventors relative to the wages offered to domestic workers for the same occupations and in the same locations. All these effects began to revert once the EAWA ban was lifted.

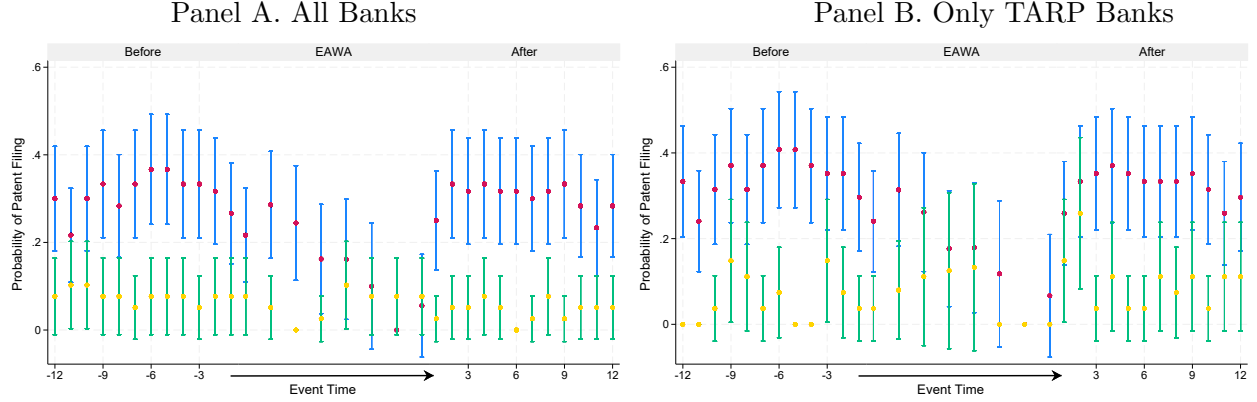
EAWA differs from the ideal experiment in several ways. First and foremost, financial institutions were not assigned to receive TARP funds at random. Moreover, during and after the 2008-2009 financial crisis, institutions were subject to major shocks to which they were differentially exposed — shocks that likely influenced their innovation activities as well as many other outcomes. And, while EAWA suddenly became binding after banks had already chosen to enter TARP, the timing of TARP exit — and, consequently, the end of the EAWA ban — was an endogenous bank-level decision.

To tackle these endogeneity concerns, we propose two empirical strategies. Our first strategy compares outcomes before and after the EAWA ban and between TARP and non-TARP recipients that had hired some foreign workers in the fields of science, technology, engineering, and mathematics (STEM workers) to capture banks that engaged in similar innovation activities before the financial crisis. This strategy can be interpreted as assessing the “extensive margin” of the EAWA ban across affected and unaffected institutions. The left panel of Figure 1 depicts the results of this assessment.⁴ Although the average

³As we elaborate in more detail below, EAWA was an ancillary piece of regulation that was approved and executed after the TARP program had started. For this reason, at the time of entering TARP, most financial institutions were not aware that a ban on new foreign hires would be imposed on them at a later date. The ban was active from EAWA’s approval in 2009 until a bank’s full repayment of TARP funds.

⁴Since we plot the mean and confidence intervals for the event times when banks filed patents, zero values indicate that

Figure 1: **Patenting Activity by U.S. Banks Around EAWA**



The figure plots the average frequency of patent filing by US banks (y-axis). The reported averages are calculated across bank-year-month observations within each event quarter (x-axis). Panel A includes both TARP and non-TARP recipients that sponsored at least one H-1B STEM job during 2004-2006. The blue (light-green) line represents TARP (non-TARP) recipients. Panel B only includes TARP recipients. The blue (light-green) line represents TARP recipients that had high (low) exposure to EAWA (see section 3.1 for how high and low exposures are constructed). The time periods are categorized as “Before,” “EAWA (Event Time),” and “After,” corresponding to quarters before, during, and after banks’ compliance with EAWA. The Event Time period for TARP recipients is institution-specific (refer to Table A.1), whereas the period of non-TARP recipients is calculated from February 17, 2009, through February 17, 2011. See section 2.1 for the selection of our sample banks. Confidence intervals are calculated at the 95% level.

patent filings of both groups of banks followed similar trends outside the EAWA period, that is, both before and after the ban was active, TARP banks (red diamonds) filed substantially fewer patents while the EAWA ban was active. By contrast, the patenting activities of non-TARP banks (yellow dots) barely changed during the same period.

The comparison of innovation trends between TARP and non-TARP banks might confound the effects of the EAWA ban with those of major economic shocks that caused TARP access by banned banks (e.g., Calomiris and Khan, 2015; Berger and Roman, 2015, 2017; Berger et al., 2020; Ma et al., 2022). To mitigate this concern, our second strategy restricts the analysis only to TARP recipients and compares the outcomes of TARP banks that, prior to the onset of the financial crisis, had a higher share of high-skilled immigrant employees relative to all high-skilled employees with TARP banks that had a lower share. We control directly for the amount of TARP funding banks received, which we interpret as a direct proxy for the severity with which banks were hit by the crisis. However, banks that relied relatively more on immigrants to fulfill their high-skilled jobs before the crisis were more affected by the EAWA ban than other banks. This second strategy can be interpreted as an assessment of the “intensive margin” of the EAWA ban within banks that were hit significantly enough by the crisis to need access to TARP.

no banks filed patents at that time, resulting in no confidence intervals.

This strategy thus only compares banks that accessed TARP and faced similarly large negative shocks during the same time period. One group relied less on immigrants to fulfill their high-skilled positions before the crisis hit, whereas the other relied more on such immigrants to fulfill the same positions.

The right panel of Figure 1 visualizes the results of our second empirical strategy. Similar to the left panel, we find that patenting trends were similar across TARP recipients that employed more immigrants among their STEM workers before the crisis (red diamonds) and TARP recipients that employed fewer immigrants among their STEM workers (yellow dots).⁵ In contrast, during the EAWA ban, patent productivity dropped substantially for TARP banks more exposed to the EAWA ban relative to less exposed TARP banks. We also verify that the profits generated by patents are not significant enough to influence the timing of banks exiting the EAWA.⁶

Our analysis is based on two main data sources. First, we obtain detailed data on H1-B visa applications from the US Citizenship and Immigration Services (USCIS). Using visa-level job classifications, we identify STEM bank employees to capture high-skill employees. Second, we compile a dataset on inventions at the level of each US financial institution using information from the US Patent and Trademark Office (USPTO). This dataset includes 3,694 bank-filed patents and 3,592 distinct inventors responsible for producing such patents. Overall, these patents carry an average market value of US\$282 million based on stock-market reactions to news about their granting.⁷ This dataset enables us to observe patents assigned to banks as well as track the employment path of bank-related inventors over time. The raw data on US banks’ innovation we construct and describe reveal that US commercial banks produce internally a substantial amount of cutting-edge innovation, especially in areas such as cybersecurity, mobile information technologies, mobile transaction technologies (e.g., robo-advising), data analytics, payment systems, FinTech applications, and several other fields that impact the efficiency, effectiveness, and security of the financial system and the overall economy.

Before proceeding with the analysis, we verify that TARP banks’ compliance with the EAWA ban

⁵Because hiring is positively correlated with bank size, we use size-adjusted cutoffs to define whether, before EAWA, a TARP bank hired more or fewer foreign STEMs relative to their peers of similar size. Specifically, we follow [Begley and Srinivasan \(2022\)](#) to classify domestic banks into small, medium, and large banks based on cutoffs of US\$10 billion, US\$50 billion, and US\$1 trillion.

⁶One possibility is that multinational banks may respond to the EAWA Act by increasing foreign affiliate employment, particularly in R&D-intensive sectors (e.g., [Glennon, 2024a,b](#)). However, the observed drop in innovation at equilibrium suggests that the offshoring of skilled labor, even when it occurs, does not fully mitigate the impact of the EAWA ban on banks’ innovation activities. In other words, considering the possibility of labor offshoring, we estimate a lower bound on the effects of this nationalist labor policy on patenting activities in the banking sector.

⁷The distribution of the private value of financial institutions from patenting is highly skewed, ranging from US\$16.9 million at the 10th percentile to US\$526.1 million at the 90th percentile of the sample distribution. For data on patent value capturing financial institutions’ private return on patenting, we employ estimates by following [Kogan et al.’s \(2017\)](#) work.

was universal; that is, applications for new H-1B visas to sponsor foreign STEM workers in domestic jobs dropped suddenly and dramatically for banks subject to the EAWA ban. By contrast, non-TARP banks with a similar level of pre-crisis foreign-STEM workers did not experience such a drop in H-1B visa applications, which dismisses the possibility that the lack of foreign workers was due to unobserved effects of the financial crisis on foreign workers' ability and willingness to migrate to the US. We then establish that during the EAWA ban, TARP banks filed patents at a considerably lower pace than non-TARP banks, with a difference of 6.5 percentage points, which is approximately 30.8% of the sample mean. These results remain robust when limiting the variation within banks over time and within year-months across banks as well as absorbing banks' foreign-hire outcomes over the preceding three years. In our second strategy, we document that TARP banks with high EAWA exposure—those that hired relatively more foreign STEM workers before the financial crisis—cut their patent filings significantly during the EAWA ban, whereas less exposed TARP banks barely cut them.

To interpret the baseline results, we first show that the drop in patenting mostly realized among patent applications submitted by first-time filers within banks, whereas patenting by existing inventors barely changed during the EAWA ban period. This finding is consistent with the fact that the EAWA ban only restricted banks from sponsoring visas for new foreign STEM hires. Moreover, we find no evidence that TARP banks subject to the EAWA ban were more likely to acquire patents from external sources during the ban, which would have allowed them to substitute in-house innovation production with externally produced patents.

We then move on to assess the labor-market implications of the EAWA ban. If the needed domestic specialized skills existed but banks hired foreign STEM workers because they were cheaper than domestic ones, and the domestic wage was not higher than the expected value of innovation accruing to banks, we should see that, after the ban, more affected banks replaced foreign human capital with domestic human capital. Note that this action should lead us to barely observe changes in patenting activities, which is not what we find. By contrast, if banks faced a shortage of specialized skills in the domestic labor market, given that existing foreign STEM workers' visas could be renewed under EAWA, we should expect wages — the price of labor for existing foreign STEM workers — to increase more than those of non-STEM workers and domestic workers. This relative increase would result from banks striving to retain the limited pool of specialized foreign STEM workers to which they had access. In this case, existing foreign workers hired before EAWA would gain bargaining power during the EAWA ban, as they had outside

opportunities both in the US (non-financial firms and non-EAWA-banned banks) and abroad.

The visa-application-level data enable us to compute the wage premium associated with each visa application filed by banks on behalf of foreign workers relative to the prevailing domestic wage for similar jobs.⁸ We find that during the EAWA ban, TARP banks offered a higher wage premium to their existing STEM foreign workers than to their existing non-STEM foreign workers, whose skills were less specialized and easier to replace domestically. And, this premium is on top of the domestic average wage for similar occupations and in the same locations, which banks need to report in the visa applications they file on behalf of foreign employees.

Overall, our empirical results appear consistent with the view that nationalistic labor policies targeting high-skilled human capital — such as banning the hiring of foreign workers in high-skill occupations — reduce domestic firms’ ability to innovate, presumably because the needed skills are too scarce in the domestic workforce.⁹ At the same time, firms affected by the ban face higher wage bills, as they must offer relatively higher wage premia to their existing pool of foreign high-skilled workers to avoid separation. We find no evidence supporting the claim that nationalistic labor policies benefit domestic high-skilled workers by providing them with more or better-paid job opportunities.

A. Related Literature

Our paper builds on several strands of literature in economics and finance. First, it relates to the literature studying the impact of policy interventions and regulation on innovation (see [He and Tian, 2020](#) for a recent review as well as [Bayazitova and Shivdasani, 2012](#); [Amore et al., 2013](#); [Agarwal et al., 2014](#); [Mayer et al., 2014](#); [Lucca et al., 2014](#); [Agarwal et al., 2017](#); [Genin et al., 2022](#); [Denes et al., 2023](#); [Babina et al., 2023](#)). Consistent with the arguments in [Lerner et al. \(2023\)](#), we find post-financial crisis regulation may have led to lower investment in innovation. Rather than studying policies intentionally designed to influence innovation, such as place-based policies (e.g., [Lerner, 1996](#); [Tian and Xu, 2022](#); [Xu, 2021](#); [D’Acunto et al., 2021, 2019](#)), our focus is on a labor-market restriction that produced unintended and yet substantial consequences for innovation production. We draw upon research documenting the unintended impacts of labor-market policy interventions on firms’ innovation and productivity, such as [Bena, Ortiz-](#)

⁸The prevailing wage in the H-1B data is visa-specific information and is self-reported by firms (visa petitioners) who provide the source of information for the prevailing wage to the immigration office. The immigration office assesses whether the reported domestic prevailing wage is plausible as part of the visa approval/denial process.

⁹Of course, the loss to the US economy may be much greater than the loss to any individual firm, given that the social benefits of the unproduced innovation would likely have exceeded the private benefits a firm could have captured.

Molina, and Simintzi (2022), who show labor protection laws prompt firms to pursue process innovation and automation rather than increase wages and job security. In the same vein, Hombert and Matray (2018) and Bena and Simintzi (2019) explore the innovation and labor-market ramifications of US free-trade agreements with China. Given the pivotal role of financial innovation for product development and transaction security within an information-driven economy, the unintended consequences of nationalistic labor policies we uncover could have far-reaching implications for US national security and position at the forefront of the global technological frontier. We share this policy implication with Canayaz et al. (2024), who show how after the US adopted protectionist policies in 2018, the employment of high-skilled workers in the chip manufacturing industry and the production of human capital specialized in that strategic industry declined dramatically.

Second, our paper is among the first to shed light on the extensive amount of innovation produced by US financial institutions. Many authors explore the causes and consequences of financial innovation at the micro level, including both patentable and non-patentable ideas (e.g., Tufano, 1989, 1995; Lerner, 2002; Tufano, 2003; Lerner, 2006, 2010; Rysman and Schuh, 2016; Pérignon and Vallée, 2017; Calvet et al., 2018; D’Acunto, 2020; Lerner et al., 2023).¹⁰ In a recent study, Chen et al. (2019) employ machine learning to identify and classify innovations in the financial industry. The authors highlight that banks emerge as the most active innovators in the financial services sector among public firms.

Third, we show the important role of high-skilled immigrants in the advancement of US FinTech, which contributes to an emerging body of literature emphasizing the contribution of domestic and foreign talent to overall US innovation and productivity (e.g., Kerr, 2008; Kerr and Lincoln, 2010; Stuen et al., 2012; Moser et al., 2014; Peri et al., 2015; Jaimovich and Siu, 2017; Dimmock et al., 2022; Bernstein et al., 2019; Burchardi et al., 2020; Arkolakis et al., 2012; D’Acunto et al., 2023; Canayaz et al., 2024). Glennon (2024b) argues that highly skilled immigrants not only improve firms’ productivity and innovation but also shape the foreign investment and trade decisions of firms, positively influencing their success abroad. Skilled immigrants shape both where and how firms undertake innovation. Our work builds upon studies employing H-1B visa data in the US, such as Bernstein et al. (2019), which delineate immigrants’ impact on US innovation, and Dimmock et al. (2022) and Chen et al. (2021), who find H-1B visa lottery winners generate more patents and patent citations than the average US domestic innovator. We also relate to Ouimet and Zarutskie (2020), who document that acquiring labor skills can be vital to firms to motivate

¹⁰Frame and White (2004), Frame and White (2012), and Lerner and Tufano (2012) provide extensive literature reviews.

their merger and acquisition activity.

2 Data

We employ several sources of data that cover information about the hiring policies and patenting activities of financial institutions.

2.1 Selecting the Sample of Financial Institutions

To obtain a homogeneous sample of financial institutions, we apply a set of filtering criteria. First, we exclude TARP participants that do not fall under the category of commercial banks, such as mutual funds, insurance companies, and trading companies. Second, we filter institutions that do not produce any innovation by keeping only institutions that submitted at least one patent application to the USPTO during our sample period (2002-2014). Third, to ensure we consider banks that had any scope for producing innovation during the EAWA period, we further exclude banks that had not filed any patents prior to December 31, 2006. These selection steps lead us to include 27 TARP recipients in our sample, whose characteristics we discuss below and report in Table A.1.

Table A.1 provides a comprehensive list of the 21 TARP recipients included in our sample, along with timelines detailing their entry into and exit from TARP. The table also includes statistics such as patent filings before 2006, foreign STEM hires during 2004-2006, total assets (millions USD) recorded as of end of 2009,¹¹ and the number of days of exposure to EAWA.

2.2 Timing of Exposure to the EAWA Foreign Hiring Ban

The EAWA came into effect on February 17, 2009, and ceased on February 17, 2011. Because financial institutions entered and exited the TARP program on different dates, exposure to the EAWA foreign hiring ban varies across institutions. Figure A.1 in the Online Appendix showcases graphically different staggered exposures to the ban across institutions.

We list the timing of the beginning and end of the EAWA ban restrictions for the TARP banks in our sample in Table A.1. The vast majority of institutions started facing the EAWA ban on the day it entered into effect, because they had already entered TARP well before the US Congress debated and approved

¹¹Initial financial data for the banks were primarily sourced from bank regulatory records and augmented with information from BvD bankscope, Compustat North American, and Global. Because Washington Mutual was acquired by J.P. Morgan on September 25 2008, the bank’s assets as of 2009 were not available. We thus use Washington Mutual’s assets in December 31, 2007.

the EAWA. For this reason, before deciding to access TARP, most institutions were unaware that they would face a foreign hiring ban, which reduces the concern that considerations regarding foreign hiring policies and innovation production might correlate with the possibility or timing at which banks started to face the ban. In the second to last column of Table A.1, we compute the number of days for which the EAWA was effective for each TARP participant. Substantial variation exists in the length of EAWA-ban exposure across TARP institutions.

2.3 Patents Data

Building on existing literature (see recent surveys by Bernstein et al., 2019; He and Tian, 2020), we measure innovation activity based on patent applications and outcomes. Our data, sourced from PatentsView, includes US patents filed by both TARP participants and other financial institutions over the period 2002–2014. The USPTO later granted these patents. Patent applications lack distinct firm-level or inventor-level identifiers for consistent tracking of assignees and inventors over time. Nevertheless, PatentsView employs a disambiguation algorithm that facilitates the association of the same inventor or patent assignee with multiple patents by pooling similar individual and firm names.

We create the sample of patents filed by financial institutions in several steps. First, we compile a list of names of institutions that filed and published at least one patent during the period 2002–2014 based on bank names in bankscope by Bureau van Dijk. Second, we use the list of institutions to search for assignee identifiers on the PatentsView platform. Third, we extract the patents filed by the banks in our sample.¹² The information we extract includes application/patent number, filing date, patent classification, assignee location, inventors’ identities and addresses, and other details. When information is missing on PatentView, we search for it directly in the USPTO patent list.

The final sample consists of 8,097 patents filed between January 2000 and December 2014, all of which were eventually granted. For these patents, we also gather citation data from Google Patents, which is current as of January 2022. Most of our empirical analysis focuses on the period of January 2007–December 2014, during which the institutions in our sample filed 3,649 patents.

Panel A of Figure 2 shows the total number of patents filed by all financial institutions in each year-quarter over the period 2002Q1–2014Q4. Similar to Lerner et al. (2023), we detect a surge in innovation production, measured by the number of filed patents that were ultimately granted, in the first decade

¹²<https://patentsview.org/download/data-download-tables>.

of the 21th century. This trend appears to have flattened since 2010. Panel B lists the top 20 financial institutions ranked by the number of patents produced from January 2007 to December 2014. The patent distribution is heavily skewed, and sizable variation in innovative activities also exists among larger banks.

Panel A of Table 1 presents the top 20 classifications of patents filed by the financial institutions. Panel B of Table 1 presents information on the assignee and abstract for the top 20 patents by citations in our dataset. We present details in section A.1.2 of the Online Appendix.

2.4 H-1B Visa Applications Data

The US H-1B Program operates on an employer-centric model; that is, employers submit applications on behalf of the prospective employees. For this reason, H-1B visa applications lack demographic details specific to the foreign worker on behalf of which they are submitted. Our H-1B visa data are sourced from USCIS and are organized at the firm-location-job-application-date level. Each visa application provides information such as the sponsoring employer’s name and location, the date on which the application was filed, and the occupation title the prospective employees would obtain. We also extract the effective start and end dates of the visa, the wages that were offered to applicants, the US prevailing wages for the same occupations in the same locations, the number of employees associated with each visa application, the designated location for employee deployment, and the application outcome—approval or rejection.

2.5 Descriptive Statistics

As discussed above, we require that the banks in our sample had filed at least one patent that was eventually granted by the USPTO prior to December 31, 2006, in order to capture banks that had scope to produce innovation during the EAWA. For the first empirical design in which we compare TARP and non-TARP banks, to exclude banks that had no scope to be subject to the EAWA ban, we require that the banks in the sample hired at least one foreign employee for a STEM job through the H-1B visa program between 2004 and 2006.¹³ Based on these criteria, the working sample includes 20 TARP and 13 non-TARP banks. Instead, in the second sample that only includes TARP banks who were all affected substantially by the 2008-2009 financial crisis shock, we have 27 banks.

Table 2 presents a set of descriptive statistics of our main analysis at the bank-year-month level. The sample period is from January 2007 through December 2014. We start in January 2007 to ensure the

¹³We follow Cheng et al. (2014) and use the years 2004–2006 to measure the pre-crisis period, but our results are similar if we consider other time periods before 2008.

outcome variables do not overlap with the pre-crisis period in which we measure banks' exposure to the EAWA shock based on their pre-crisis hiring of foreign STEM workers, which is 2004–2006, as we define below. In Panel A of Table 2, we consider the sample in which we implement our first empirical strategy. According to this sample, approximately 60% of observations are contributed by TARP recipients and 9.4% of bank-year-month observations were subject to the EAWA ban. In addition, the frequency with which banks hired at least one H-1B STEM employee is 37.6%. The average number of foreign STEM workers hired per bank-year-month is seven. Furthermore, 20.5% (4.8%) of bank-year-months involve at least one patent filing (being purchased), 14.8% are related to at least one patent filed by inventors who filed patents for the bank for the first time, 8.5% are related to patents filed by first-time filers who are lead inventors, and 5.8% observations pertain to year-quarters in which all patents are filed by non-first-time filers. Over the past three years, 84% (75.3%) of observations correspond to banks that hired at least one H-1B-sponsored worker (H-1B STEM).

Panel B shifts focus to the sample used for our second empirical approach. Here, the TARP banks' average funding from the government was 2.5% of their total assets as of the end of 2009. Within this sample, 15.9% of the bank-year-month observations fell under the EAWA ban. The proportion of observations with at least one H-1B STEM worker hired drops slightly to 36.1%. The mean number of foreign STEM employees hired per bank-year-month rises marginally to 7.4. In addition, 24.7% (6.1%) of the bank-year-months involve at least one patent filing (being purchased). The share of first-time patent filers who are lead inventors is 10.2%, while 17.7% of observations relate to patents filed by inventors filing for the first time. Lastly, observations where all patents are filed by non-first-time filers account for 7%. Over the previous three years, 67.8% (63.3%) of observations are linked to banks that have hired at least one H-1B-sponsored worker (H-1B-sponsored STEM).

3 Empirical Strategies

In this section, we discuss our two difference-in-differences (DiD) strategies and provide evidence about the compliance of TARP recipients with the EAWA ban on hiring foreign workers until the full repayment of TARP funds.

3.1 First Difference-in-Differences Strategy: TARP vs. Non-TARP Banks

We implement the first empirical strategy by estimating a set of linear specifications that exploit within-bank variation in both dependent and independent variables. The most complete specification is as follows:

$$\begin{aligned} Outcome_{i,s} = & \alpha + \beta_1 EAWA_{i,s} + \beta_2 EAWA_{i,s} \times TARP_i + \beta_3 Post_{i,s} + \\ & \beta_4 Post_{i,s} \times TARP_i + X' \times \beta_5 + \eta_i + \eta_s + \epsilon_{i,s}, \end{aligned} \quad (3.1)$$

where $EAWA_{i,s}$ is a dummy variable that equals 1 if institution i is subject to the EAWA ban on hiring foreign workers during year-month s , and zero otherwise. $TARP_i$ is a dummy variable that equals 1 if institution i accessed the TARP program, and 0 otherwise. $Post_{i,s}$ is a dummy variable that equals 1 during year-month s after the EAWA is no longer binding for i , and 0 otherwise. $EAWA_{i,s}$ and $Post_{i,s}$ are set to 0 for institutions that never borrowed funds through TARP.¹⁴ We also incorporate full sets of institution- and year-month fixed effects (η_i and η_s). The inclusion of bank fixed effects absorbs systematic and time-invariant cross-sectional differences across banks. Year-month fixed effects allow us to absorb economy-wide aggregate and time-varying shocks that affect all banks uniformly. Given the span of our sample period, covering the financial crisis and the Great Recession, unobservable time-varying aggregate shocks are likely to be important drivers of banks' outcomes, including their innovation activities. $Outcome_{i,s}$ are a set of outcomes measured at the level of institution i in year-month s .

In equation (3.1), the coefficient of interest is β_2 , which captures the within-bank change in outcomes during the EAWA period relative to before EAWA entered into effect and across TARP and non-TARP recipients.

3.2 Second Difference-in-Differences Strategy: Varying Pre-Exposure to EAWA among TARP Banks

An important concern with our first empirical strategy is that the coefficient β_2 in equation (3.1) might capture the effect of time-varying bank-specific shocks that affect TARP and non-TARP institutions in a systematically different way. For instance, if TARP and non-TARP recipients' innovation activities were affected differently by monetary policy interventions during the period of interest or if TARP banks were exposed to shocks that instead did not affect non-TARP banks during the period of interest, the

¹⁴Because only TARP banks were subject to the EAWA restriction, the two variables $EAWA_{i,s}$ and $EAWA_{i,s} \times TARP_i$ are perfectly correlated. For this reason, we only report the estimates of $EAWA_{i,s} \times TARP_i$ in all tables. The same logic applies to the variables of $Post_{i,s}$ and $Post_{i,s} \times TARP_i$.

coefficient of interest in equation (3.1) might capture such differences.

To address this concern, our second empirical strategy focuses only on TARP recipients, which dismisses by construction the concern that TARP and non-TARP recipients might have faced different time-varying shocks that affected their innovation activities during the EAWA ban.

Within this subset of institutions, we compare innovation outcomes across banks that had different pre-determined exposure to EAWA before EAWA was approved and entered into effect. We capture exposure with the share of bank-level STEM workers who were foreign in the pre-crisis period. The rationale for this strategy is that irrespective of whether banks might have been able to foresee the financial crisis, the implementation of a ban on foreign worker hiring was barely foreseeable in the pre-crisis period. Indeed, the EAWA ban was largely the result of political negotiations in the US Congress to secure the number of votes needed to pass the broader set of subsidies and other financial aid to institutions during the financial crisis, including subsequent waves of financing of TARP as well as other funds/programs. For this reason, banks could not have chosen the share of foreign workers among their STEM workforce in the pre-crisis period based on their expectations about the future EAWA ban. At the same time, once the ban was implemented, banks that relied more than others on foreign STEM workers to fill their STEM job openings were affected more than other banks.

To implement this strategy, we face the concern that both the distribution of bank sizes and of patenting activities are heavily skewed. Suppose, for instance, that a small bank only employs two STEM workers and one of them is a foreign national. The share of foreign STEM workers would be 50% for this bank. This bank would appear highly exposed to EAWA, but at the same time, its patenting activities would be very low and volatile irrespective of the EAWA ban. By contrast, a very large bank that produces many patents regularly might hire hundreds of STEM workers, and the fraction of foreign workers among them is likely lower than 50% given the frictions and costs firms face to hire foreign workers. If we were capturing exposure to the EAWA ban with the raw ratio of foreign STEM workers, we would by construction attribute high exposure to a group of banks whose patenting activities are not comparable to those of the largest innovators, and vice versa.

To tackle this issue, we follow [Begley and Srinivasan \(2022\)](#) and classify domestic banks into small, medium, and large banks. Small banks have assets below US\$10 billion, medium banks have assets between US\$10 and US\$50 billion, and large banks have assets above US\$1 trillion. Within each of the three groups, we then compute a dummy variable, *High Exposure_i*, that equals 1 if the number of foreign

STEM workers bank i hired in 2004-2006 was above the median of the distribution of foreign STEM workers within the group of banks of similar size. The median cutoffs for the number of foreign STEM hires by small, medium, and large banks are 0, 1, and 10.5, respectively. In this way, the categorization of TARP banks into different levels of exposure to EAWA is conditional on banks' size.

Based on these considerations, we implement our second strategy by estimating the following DiD specification:

$$\begin{aligned} Outcome_{i,s} = & \alpha + \beta_1 EAWA_{i,s} + \beta_2 EAWA_{i,s} \times High\ Exposure_i + \beta_3 Post_{i,s} + \\ & \beta_4 Post_{i,s} \times High\ Exposure_i + X' \times \beta_5 + \eta_i + \eta_s + \epsilon_{i,s}. \end{aligned} \quad (3.2)$$

One concern underlying this strategy is that banks' reliance on STEM human capital for operations other than innovation activities may be correlated with the deterioration of financial conditions during the crisis. For instance, recent studies show firms tend to lose talented workers, which might include STEM workers who innovate as well as STEM workers in other occupations, as they approach bankruptcy (e.g., Baghai et al., 2021, 2023). In fact, the shortage of high-skilled immigrants and financial constraints may represent two distinct channels that force TARP banks to cut their innovation activities.

To fully control for the bank-specific change in financial conditions during the crisis, we gathered data on the upfront funding obtained by each TARP recipient. We construct a continuous variable ($TARP\ Fund\%_i$) calculated as the proportion of (troubled) assets purchased by the US Department of the Treasury from institution i , scaled by i 's total assets as of 2009.¹⁵ We consider this measure to be the most direct proxy of the full extent of bank-level observed and unobserved deterioration of financial conditions during the crisis because, given the physical and reputational costs of accessing TARP funds, banks had the incentive to only access them to the minimal extent needed to weather the crisis. Other balance-sheet-level proxies would not provide such direct information on the deterioration of each bank's financial conditions during the crisis.

¹⁵The data on troubled assets (e.g., toxic assets and equity) purchased by the Treasury were obtained from the "TARP Investment Program Transaction Reports," available at: <https://home.treasury.gov/data/troubled-assets-relief-program/reports/tarp-investment-program-transaction-reports>. Due to acquisitions or mergers involving some TARP participants, individual funding data were not available for acquired banks in the following cases: (i) Bank One Corporation acquisition by JPMorgan on July 1, 2004; (ii) Merrill Lynch acquisition by Bank of America in September 2009; (3) Wachovia Corporation acquisition by Wells Fargo on December 31, 2008; (4) Washington Mutual acquisition by JPMorgan Chase by the end of 2009; and (5) Chase Manhattan Bank acquisition by JPMorgan Chase on December 31, 2000. In such cases, we utilized the TARP funding data of their respective acquirers.

4 Foreign Hiring Restrictions and Innovation

We now move on to assess the effects of the EAWA ban on banks' innovation production. First, we confirm that TARP recipients complied with the EAWA ban on hiring foreign workers; that is, the EAWA was effectively enforced and banks stopped hiring new foreign workers while subject to it. Second, we report the results for implementing the empirical strategies discussed above.

4.1 Compliance with EAWA: New Foreign Workers Hiring by TARP Banks

Panel A of Figure 3 contrasts the quarterly average number of H-1B STEM workers hired by the TARP banks with those hired by the non-TARP banks in our sample. When EAWA was passed, TARP banks experienced a large and abrupt drop in foreign STEM hires, but they gradually returned to the pre-EAWA hiring levels after EAWA ended. By contrast, non-TARP banks' foreign hiring was similar across different time periods. Panel B of Figure 3 contrasts the TARP banks that had employed more foreign STEM workers during 2004-2006 with similarly troubled TARP banks that had employed fewer foreign STEM workers during the same period. Notably, TARP banks with lower EAWA exposure barely hired any foreign STEM workers before, during, and after the EAWA restriction, which is why the U-shaped curve of foreign STEM hires was driven by TARP banks.

In Panels C and D of Figure 3, we consider the quarterly frequency at which banks sponsored at least one H-1B STEM visa. Again, we can see that TARP banks cut the likelihood of hiring in response to their compliance with EAWA and, relative to TARP banks that were less exposed to EAWA, more exposed TARP banks halted their hiring of foreign STEMs during the EAWA ban.

To be consistent with the subsequent analysis, we perform this "first-stage" assessment more formally by estimating our linear specifications. The first outcome variable we consider (see Panel A of Table 3) is the natural logarithm of 0.001 plus $\#STEM_{i,s}$. $\#STEM_{i,s}$ is the number of foreign STEM workers sponsored by bank i in year-month s .¹⁶ For robustness, in Panel B of Table 3, we consider an alternative dependent variable, $STEM\ H1B_{i,s}$, which is a dummy variable that equals to 1 if bank i sponsored at least one H-1B visa in s , and 0 otherwise.

For each panel of Table 3, columns (1)-(4) report the results for estimating equation (3.1), which is based on the full sample of banks, while columns (5)-(8) for estimating equation (3.2), which only

¹⁶We are not concerned about using a linear specifications when the outcome is the logarithm of a count variable in this case (e.g., Cohn et al., 2022), because this analysis does not aim to interpret the economic magnitudes of the coefficient estimates but to verify that, indeed, the hiring of H-1B workers dropped to zero during the EAWA ban for affected banks

includes TARP banks. Columns (1)-(4) of Panel A of Table 3 reveal that during the EAWA ban, TARP banks cut foreign STEM hires more than non-TARP banks did. Columns (1)-(4) of Panel B quantify this difference showing that TARP banks were 14-24 percentage points less likely to hire foreign STEM workers relative to non-TARP banks. This number is about 37%-64% of the sample mean (0.376, as indicated in Table 2) and 29%-50% of a standard deviation (0.484, as indicated in Table 2). Moving on to the within-TARP-banks analysis (columns (5)-(8)), Panel A shows a stronger EAWA effect on TARP banks that had employed more foreign STEM workers in 2004-2006. Panel B quantifies this effect. In column (8), for example, TARP banks with high EAWA exposure were 20.8 percentage points less likely to sponsor H-1B STEM visas during EAWA, which is about 57.6% of the sample mean sponsorship.

4.2 Ban on New Foreign Workers Hiring and Banks' Innovation

Moving on to assess the effects of the EAWA ban on innovation, we estimate equation (3.1) using $Patent\ Filed_{i,s}$ as the dependent variable. $Patent\ Filed_{i,s}$ is a dummy variable that equals 1 if bank i filed at least one patent in year-month s , and 0 otherwise. The specification implies a linear probability model for the likelihood of patenting, which we prefer to non-linear estimators due to the need to include a large set of bank- and time-fixed effects.

We report the results in Table 4. Columns (1)-(4) report the impact of EAWA on patenting for TARP recipients relative to non-TARP recipients, both of which had sponsored some H-1B STEM jobs in 2004-2006. In the most restrictive specification (column (4)), we find a 5.9-percentage-point decrease in the likelihood of TARP recipients filing patents. In terms of economic magnitude, this estimate accounts for 28.8% of the sample mean (0.205, as indicated in Table 2) and about 14.6% of a standard deviation of the patenting dummy (0.404, as indicated in Table 2).

Columns (5)-(8) report the results when estimating our second empirical strategy (equation (3.2)). TARP recipients with higher pre-crisis EAWA exposure — banks that hired more foreign STEMs than the size-adjusted sample median in the pre-crisis period — experienced a significant drop in patent filing relative to otherwise similar TARP recipients with a lower EAWA exposure. In terms of economic significance, moving from banks for which *High Exposure* equals 0 to those for which it equals 1 leads to a 10.1-percentage-point drop in patent filings. This number represents 40.9% of the sample mean likelihood of patent filing in a year-month (0.247, as indicated in Table 2) and 23.4% of a standard deviation of the patenting dummy (0.432, as indicated in Table 2).

Beyond the baseline results, we assess if exposure to the EAWA ban affected patenting activities differently or similarly across patenting areas. Specifically, we consider business-method patents (classified as US patent class 705), non-business-method patents (classified as US patent classes other than 705), and FinTech patents filed by bank i in year-month s . Details about the keyword list we use to classify patent filings across areas are in Table A.2 of the Online Appendix.¹⁷ Our analysis identifies 1,312 FinTech patents filed from 2007 to 2014, with Figure A.2 in the Online Appendix showcasing the top 20 financial institutions in terms of FinTech patent filings. Because a patent may be classified as both FinTech and business method to address potential overlap we label all business-method patents as non-FinTech, regardless of the keywords.

Table 5 reports the results. In columns (1)-(3), we can see that the patenting drop for TARP banks during the EAWA ban appears stronger for business-method patents even though statistical significance is sparse. When moving to the within-TARP-recipient analysis in columns (4)-(6), we detect a more uniform drop in patenting across areas in terms of both magnitudes and lack of statistically significant differences.

4.3 Innovation Quality: Patent Citations

The results so far referred to the extensive margin of innovation production, i.e. the quantity of new patents filed. The intensive margin of innovation, namely, the importance or quality of the innovation in terms of generating future follow-up ideas and disrupting product markets and processes, is also important. On the one hand, firms might respond to human capital restrictions by focusing their innovation activities to produce fewer but individually more important patents, thus concentrating the limited resources available for patent production into higher-return projects. On the other hand, banks could focus on easier-to-produce but less disruptive innovation, which would further aggravate the overall domestic drop in cutting-edge innovation as a consequence of bans in hiring high-skill immigrants.

To capture the intensive margin of innovation, we use patent citations as a standard proxy for patent quality that we can measure in our data. Figure A.3 of the Online Appendix plots the distribution of citations for the patents in our sample. For this analysis, we cannot use the bank-year-month level sample

¹⁷Inspired by Chen et al. (2019), we employ a keyword-based approach, examining seven classes: cybersecurity, mobile transaction, data analytics, blockchain, peer-to-peer lending, robo-advising, and the internet of things. We classify patents based on the presence of keywords in their title, abstract, and claims. For example, the term “authentication” defines the keyword for “cybersecurity.” If a patent contains the keyword “authentication” in its title, abstract, or claims, it is categorized as related to cybersecurity.

that also includes periods during which banks submitted no granted patents, which would be vacuously associated with zero citations. Instead, we restrict the sample to observations in which at least one patent filed was ultimately granted. Our dependent variable is $\overline{Adj.Cites}_{i,s}$, which is the average of $Adj.Cites$ for all granted patents submitted by bank i in year-month s . Following [Bernstein et al. \(2019\)](#), we determine $Adj.Cites$ by normalizing the number of citation counts based on the average citations within a specific technology-class-year. We transform $\overline{Adj.Cites}$ into a percentile-rank form.¹⁸ Table A.3 of the Online Appendix reports a set of summary statistics for our measures of patent citations.

Table 6 reports the results for our two strategies. Similar to the extensive margin of innovation in Table 5, we also include separate estimates for patents in different technological areas. As shown in columns (1)-(4), when comparing patent quality between TARP and non-TARP recipients, we fail to detect any systematic differences in the citations of patents filed during the EAWA period by TARP and non-TARP banks.

In contrast, we estimate an overall positive coefficient for the full sample of patents when comparing citations within TARP banks that were differently exposed to the EAWA ban. Despite being positive and statistically significant, the estimated coefficient (0.085 in column (5)) is small. We find that it masks dramatic heterogeneity across patent categories. In fact, during the EAWA ban, more exposed TARP recipients produced business-method patents with higher average citation counts while at the same time producing non-business-method and FinTech patents that had lower average citation counts. Although our setting does not allow for a direct test, these results might suggest that when banks face restrictions to the hiring of inventors, they might refocus their innovation to develop high-quality cost-cutting methods to manage existing processes, i.e. high-quality business-method patents. At the same time, when unconstrained, banks might focus their innovation activities on developing new disruptive products and future commercial opportunities, which would fall under the FinTech/non-business-method patent categories.

¹⁸We aggregate variables at the new level of observation, that is, bank-year-month observations associated with at least one filed patent eventually granted. Here, the year corresponds to the filing year of all patents belonging to the same technology class. Our citation data are observed up until January 2022. Figure A.3 of the Online Appendix plots the distribution of citation counts for patents submitted by institutions in our sample spanning the years 2007–2014. As anticipated, our findings reveal a pronounced positive skewness in the distributions of both raw and adjusted citation counts.

5 Interpreting the Effects of Banning New Foreign Hires on Innovation

In this section, we provide additional tests and evidence that help us interpret the economic channels that drive the baseline results.

5.1 Are New Hires So Relevant for Innovation? Why?

As we discussed in the institutional setting, EAWA imposed restrictions exclusively on the sponsorship of *new* H-1B visas, whereas the renewal of H-1B visas for foreign workers who were already employed before EAWA remained unaffected. In other words, EAWA did not affect the existing stock of human capital provided by foreign innovators that banks already employed before the ban. But then, are new hires important for innovation activities relative to existing innovators?

We provide evidence that they are. Indeed, the results in Table 7 collectively suggest that the overall decline in patenting due to EAWA exposure is largely driven by the disappearance of patent filings in which new inventors were part of the filing team. Specifically, we estimate our baseline specifications defining $First\ Filer_{i,s}$ as the outcome variable, which is a dummy variable that equals 1 if, in year-month s , at least one patent was filed by an inventor who submitted patents for the first time during their tenure at bank i . This dummy variable captures the patents filed by inventors who were present for the first time in banks' filing teams, which are exactly the ones for whom the EAWA ban was relevant.

We report the results in Panel A of Table 7, which reveals a consistent pattern irrespective of whether we compare $First\ Filer$ between TARP and non-TARP banks or within TARP banks with varying EAWA exposures. In both cases, banks subject to the ban experienced a larger drop in the likelihood of filing patents for which new inventors were part of the patenting team. Importantly, the estimated effects closely align with our baseline estimates for the likelihood of banks filing any patents (see Table 4).

In Panel B, we again estimate our baseline specifications, where $Non-First\ Filer$ is the outcome variable, which is a dummy variable that equals 1 if, in year-month s , all patents filed by bank i are not filed by any first-time filers for this bank. Across all columns, banks subject to the EAWA ban experienced *no* drop in the likelihood of filing patents for which the entire patenting team consists of non-first filers.

After documenting that new inventors are indeed crucial drivers of our results, the question remains

to why this is the case. One possibility is that banks hire inventors who have already produced a patentable innovation but, facing the costs and uncertainties of the patenting process, might prefer to sign an agreement with an employer that purchases the ownership rights over the invention in exchange for a full-time employment job offer (and, potentially, other payments and benefits).

We are not aware of any systematic data sources that would enable us to observe whether newly hired inventors bring pre-existing unpatented innovative ideas to the banks that hire them. In the absence of such data, we proxy these instances with patent applications filed by financial institutions wherein an inventor who files for the first time during year-month s of their tenure with bank i is not only part of the team but also serves as the lead inventor; that is, s/he is listed as the first name in the inventors' team. The rationale is that lead inventors typically play a significant role in driving the innovation the patent protects. Therefore, when lead inventors are newly hired employees, the likelihood that the new hires contributed their existing innovation ideas to the bank once recruited is higher.¹⁹ Consistent with this possibility, in Table A.4 of the Online Appendix estimates the effect of EAWA on patenting conducted by new inventors who are also listed as the lead inventor. We find a substantial share of the patents that EAWA-exposed banks did not file during the EAWA period were patents for which new employees would have been part of the team as lead inventors.

5.2 Did Banks Substitute Internal Patents with External Patent Purchases?

EAWA-exposed banks could have reacted to the drop in internally-produced patents during the EAWA ban by turning to the secondary market to purchase patents developed by others, which are not observed in our sample. Indeed, an emerging body of research shows that firms frequently acquire innovation from counterparts that are more efficient at innovating (e.g., Aghion and Tirole, 1994; Bowen III, 2016; Cunningham et al., 2021; Sevilir and Tian, 2023; Puri et al., 2024). In this case, although the EAWA ban would have still forced banks to change their innovation strategies, the overall effect on US innovation production and implementation would have been more muted.

To assess this possibility, we compile the patent transaction histories of each bank in our sample using the Patent Assignment Dataset sourced from the USPTO. This dataset enables parties to record inter-firm patent assignments, ensuring a comprehensive record of claimed interests in a given patent.²⁰

¹⁹To identify a lead inventor, we do not consider patents that list inventors alphabetically, in which the order of names does not indicate the importance of the specific inventor in developing the patent.

²⁰The data are available for download on the official website: <https://www.uspto.gov/ip-policy/economic-research/research-datasets/patent-assignment-dataset>.

To precisely measure inter-firm patent purchases, we exclude inter-bank assignments caused by banks' acquisition of collateralized patents and patents from acquired firms. The summary statistics presented in Table 2 indicate the monthly frequency with which banks acquire at least one patent from external parties is 4.9% for all banks and 6.2% for TARP banks, which suggests this potential channel is not economically negligible.

Table 8 presents the estimation results based on our two proposed empirical strategies. The dependent variable is $Purchased_{i,s}$, which is a dummy that equals 1 if bank i acquired at least one patent from external parties in year-month s , and 0 otherwise. Not only are the results inconsistent with the possibility that EAWA-exposed banks replaced internal patenting production with higher purchases of external patents, but, in fact, the likelihood of external purchases declined *more* for EAWA-exposed banks than for other banks while the ban was in effect. This complementarity between internally- and externally-produced innovation by US banks, if anything, suggests that bans on hiring foreign inventors might have far-reaching implications on innovation in the financial sector above and beyond the direct effects on patent production we document in our baseline results.

5.3 Labor-Market Implications: Wage Premium to Foreign STEM Workers

In the final two sets of analyses, we assess the labor-market implications of the EAWA ban. In particular, because EAWA banned TARP banks from hiring new foreign workers, and because foreign workers in the US are not allowed to carry H-1B visas from one employer to the other, TARP recipients could only employ high-skilled immigrants by retaining their existing H-1B-visa employees, that is, the foreign workers they had hired before the crisis. The EAWA ban thus shifted the bargaining power between TARP banks and existing foreign workers in favor of workers. For this reason, the EAWA ban might have resulted in TARP banks not only facing a drop in innovation activities but also in having to pay a higher wage premium to existing foreign workers relative to non-TARP banks if they wanted to retain foreign innovators among their employees.

Based on these considerations, we analyze the wage premia foreign STEM workers obtained during and after the EAWA ban by different types of banks.²¹

We exploit information about wage premia (i.e., the difference between the wage offered to the visa

²¹Shifts in bargaining power between existing immigrants and bank employers might also be reflected in changing labor-contract covenants such as non-compete agreements and equity-based compensation (e.g., Jeffers, 2024; D'Acunto et al., 2021). Unfortunately, because these covenants are unobserved in H-1B visas, we cannot study them in our setting.

holder and the prevailing wage for the same occupation and location at the time of the visa application), job location and classification, and the number of immigrants sponsored in each petition from the H-1B visa data. From the sample from January 2007 through December 2014, we have collected a total of 50,545 H-1B petitions filed by our sample banks. The prevailing wage rate is petition-specific and is defined as the average wage paid to similarly employed workers in the requested occupation in the area of intended employment.²² Panel A of Table 9 reports summary statistics for the wage premium and other crucial covariates in this sample.

We estimate the following regression specification:

$$\begin{aligned} \overline{\text{Wage Premium}}_{i,l,s} = & \alpha + \beta_1 \text{EAWA}_{i,s} \times \text{TARP}_i \times \text{STEM}_{i,l,s} + \beta_2 \text{EAWA}_{i,s} \times \text{TARP}_i + \beta_3 \text{EAWA}_{i,s} \times \text{STEM}_{i,l,s} \\ & + \beta_4 \text{EAWA}_{i,s} + \beta_5 \text{Post}_{i,s} \times \text{TARP}_i \times \text{STEM}_{i,l,s} + \beta_6 \text{Post}_{i,s} \times \text{TARP}_i + \beta_7 \text{Post}_{i,s} \times \text{STEM}_{i,l,s} + \\ & \beta_8 \text{Post}_{i,s} + \beta_9 \text{STEM}_{i,l,s} \times \text{TARP}_i + \beta_{10} \text{STEM}_{i,l,s} + \beta_{11} \text{TARP}_i + \eta_l + \eta_i + \eta_s + \epsilon_{i,l,s}, \end{aligned} \quad (5.1)$$

where $\overline{\text{Wage Premium}}_{i,l,s}$ is the weighted-average wage premium over the prevailing wage for H-1B visas within the same job category l , submitted by bank i in year-month s . The weight is determined by the number of immigrants sponsored under each H-1B visa. $\text{STEM}_{i,l,s}$ is a dummy variable that equals 1 if job category l for visas sponsored by bank i in year-month s falls within the STEM classification, and 0 otherwise.

Panel B of Table 9 reports the estimates. In columns (1)-(5), we detect a pattern consistent with TARP recipients paying a higher wage premium to existing STEM foreign employees during the EAWA period. In contrast, the wage premium TARP banks paid to foreign STEM workers after the EAWA ban was lifted did not differ from the wage premium they paid to others. The economic magnitude of these findings is illustrated in the last column: while TARP recipients paid foreign non-STEM workers (an insignificant) 3.5 percentage points less than the prevailing wage during EAWA, they compensated existing foreign STEM jobs with a 5.6-percentage-point higher premium.²³

²²The prevailing wage rate is usually obtained by contacting either the State Workforce Agency (SWA) or from other legitimate sources of information.

²³In Panel B of Table 9, *EAWA* and *Post* are absorbed because their perfect correlation with $\text{EAWA} \times \text{TARP}$ and $\text{Post} \times \text{TARP}$, respectively. For the same reason, $\text{EAWA} \times \text{STEM}$ and $\text{Post} \times \text{STEM}$ are absorbed because their perfect correlation with $\text{EAWA} \times \text{TARP} \times \text{STEM}$ and $\text{Post} \times \text{TARP} \times \text{STEM}$, respectively.

5.4 An Anatomy of STEM Jobs' Skill and Knowledge Sets

Our findings, considering both patenting activities and labor-market implications, imply that STEM immigrants could potentially play a more significant role in driving innovation within financial institutions than non-STEM immigrants. To investigate this possibility more directly, we conduct a comparative analysis of the skill and knowledge profiles of STEM jobs with those of non-STEM jobs. Both STEM and non-STEM positions in the US are sponsored under the H-1B visa program, which was subject to the EAWA ban for TARP fund recipients.

We obtain information on skills at the occupation level from O*NET. Within the Standard Occupational Classification (SOC) system, each occupation's O*NET rating quantifies the extent to which a specific skill or knowledge component is unique to that occupation. Skill components include basic skills (e.g., reading, facilitating the acquisition of new knowledge) and cross-functional skills (e.g., problem-solving, spanning across various domains of activities). These skills are evaluated across diverse areas such as business and management, manufacturing and production, engineering and technology, mathematics and science, health services, and others.²⁴

Drawing from these occupation-level ratings, we measure the differences in skill and knowledge between foreign-born employees in STEM and non-STEM occupations. More precisely, we employ numerical ratings to quantify the level of a descriptor (a skill or knowledge component) h to N STEM occupations relative to the level of the same descriptor to M non-STEM occupations. For an intra-bank comparison, both types of occupations must be sponsored by financial institutions i in the 2004-2006 period. We compute the following:

$$Difference_{h,i} = \frac{\frac{\sum_{k=1}^N \sum_{l=1}^n Rating_{h,k}}{\sum_N \sum_n 1}}{\frac{\sum_{k=1}^M \sum_{m=1}^m Rating_{h,k}}{\sum_M \sum_m 1}} - 1, \quad (5.2)$$

where $Rating_{h,k}$ represents the rating assigned to the level corresponding to descriptor h for occupation k . Financial institution i sponsors n H-1B workers for each STEM occupation and m H-1B workers for each non-STEM occupation. Using equation (5.2), we compute the average of $Difference_{h,i}$ across J financial institutions to determine the rating of descriptor h for STEM occupations relative to non-STEM occupations:

$$Difference_h = \frac{\sum_J Difference_{h,i}}{J}, \quad (5.3)$$

²⁴To synchronize with the timeline of STEM-dependence that we measure, we obtained O*NET version 11.0 as of December 2006 from O*NET.

where J is the number of financial institutions sponsoring H-1B jobs in our sample.

Our null hypothesis posits that ratings for the skill component h are consistent across various occupations, and as a result, $Difference_h$ in equation (5.2) is 0. Panel A of Figure 4 illustrates $Difference_h$ calculated using skill ratings across 20 skill elements. Note skills such as “Science,” “repairing,” “installation,” “equipment maintenance,” and “programming” emerge as particularly advantageous for an average STEM occupation. In Panel B of Figure 4, t-statistics for the mean, computed according to equation (5.3), are plotted, revealing that the majority of skill differences between STEM and non-STEM occupations are statistically significant. Moving to Panel C of Figure 4, $Difference_h$ is calculated using knowledge ratings across 20 elements. Similar to the skill ratings, “Science,” “repairing,” “installation,” “equipment maintenance,” and “programming” are identified as advantageous skill sets for an average STEM occupation. In Panel D, $Difference_h$ is constructed using knowledge ratings across 33 elements. Here, STEM occupations exhibit a performance exceeding 200% in various knowledge areas, including “biology,” “physics,” “chemistry,” “fine arts,” and “design.” However, these differences are only marginally statistically significant.

Overall, our findings indicate STEM and non-STEM workers contribute significantly distinct skill sets to the financial institutions that employ them. As a result, these workers are not readily interchangeable.

6 Conclusions

Nationalistic labor policies that restrict domestic employers’ ability to hire specialized foreign workers can significantly impede domestic innovation production and quality even in strategic and frontier areas such as FinTech and cybersecurity.²⁵

Our results suggest a few paths for future research. First, assessing and estimating the effects of lower patenting activities by financial companies that cannot hire specialized foreign STEM workers in terms of investment and profitability in the long run would be important to quantify fully the magnitude of the negative effects of nationalistic labor policies in this area. Moreover, a premise of our analysis is the study of an area of the labor market in which skills are highly specialized and scarce. Whether domestic firms would react to bans on foreign hiring in less specialized occupations by hiring more domestic workers at higher wages can only be assessed in a different institutional setting.

²⁵Berg et al. (2022) review the growing literature on FinTech lending. The authors argue that increased convenience and speed are the primary drivers of the growth of FinTech lending.

Overall, the growing global threat of cyberattacks against financial corporations, along with other risks to the security of national financial sectors, emphasizes the national security implications of banks' innovation activities. Losing the technological edge in this domain could have far-reaching negative implications above and beyond the short- and medium-term economic effects on banks and employees. Assessing the magnitude of these effects is crucial for informing regulators about the potential unintended consequences of their nationalistic policies.

References

- Agarwal, S., G. Amromin, I. Ben-David, S. Chomsisengphet, P. Tomasz, and A. Seru (2017). Policy intervention in debt renegotiation: Evidence from home affordable modification program. *Journal of Political Economy* 125, 654–712.
- Agarwal, S., D. Lucca, A. Seru, and F. Trebbi (2014). Inconsistent regulators: Evidence from banking. *The Quarterly Journal of Economics* 129, 889–938.
- Aghion, P. and P. Howitt (1992). A model of growth through creative destruction. *The Review of Economic Studies* 60(2), 323–351.
- Aghion, P. and J. Tirole (1994). On the management of innovation. *The Quarterly Journal of Economics* 109(4), 1185–1209.
- Akcigit, U., S. Caicedo, E. Miguelez, S. Stantcheva, and V. Sterzi (2018). Dancing with the stars: Innovation through interactions. *NBER Working Paper* 24466.
- Alesina, A., A. Miano, and S. Stantcheva (2023). Immigration and redistribution. *The Review of Economic Studies* 90(1), 1–39.
- Amore, M. D., C. Schneider, and A. Žaldokas (2013). Credit supply and corporate innovation. *Journal of Financial Economics* 109(3), 835–855.
- Arkolakis, C., M. Peters, and S. K. Lee (2012). European immigrants and the United States’ rise to the technological frontier. *Working Paper*.
- Babina, T., A. X. He, S. T. Howell, E. R. Perlman, and J. Staudt (2023). Cutting the innovation engine: How federal funding shocks affect university patenting, entrepreneurship, and publications. *The Quarterly Journal of Economics* 138(2), 895–954.
- Baghai, R. P., R. C. Sila, V. Thell, and V. Vig (2021). Talent in distressed firms: Investigating the labor costs of financial distress. *The Journal of Finance* 76(6), 2907–2961.
- Baghai, R. P., C. Silva, Rui, and L. Ye (2023). Teams and bankruptcy. *Working Paper*.
- Bayazitova, D. and A. Shivdasani (2012). Assessing TARP. *The Review of Financial Studies* 25(2), 377–407.
- Begley, T. A. and K. Srinivasan (2022). Small bank lending in the era of fintech and shadow banks: A sideshow? *The Review of Financial Studies* 35(11), 4948–4984.
- Bena, J., H. Ortiz-Molina, and E. Simintzi (2022). Shielding firm value: Employment protection and process innovation. *Journal of Financial Economics* 146(2), 637–664.
- Bena, J. and E. Simintzi (2019). Machines could not compete with Chinese labor: Evidence from US firms’ innovation. *Working Paper*.
- Berg, T., A. Fuster, and M. Puri (2022). Fintech lending. *Annual Review of Financial Economics* 14, 187–207.
- Berger, A. N. and R. A. Roman (2015). Did TARP banks get competitive advantages? *Journal of Financial and Quantitative Analysis* 50(6), 1199–1236.
- Berger, A. N. and R. A. Roman (2017). Did saving Wall Street really save main street? The real effects of tarp on local economic conditions. *Journal of Financial and Quantitative Analysis* 52(5), 1827–1867.
- Berger, A. N., R. A. Roman, and J. Sedunov (2020). Did TARP reduce or increase systemic risk? The effects of government aid on financial system stability. *Journal of Financial Intermediation* 43, 100810.

- Bernstein, S., R. Diamond, T. McQuade, and B. Pousada (2019). The contribution of high-skilled immigrants to innovation in the united states. *Stanford University GSB. Working Paper*.
- Borjas, G. J. and K. B. Doran (2012). The collapse of the Soviet Union and the productivity of American mathematicians. *The Quarterly Journal of Economics* 127(3), 1143–1203.
- Bowen III, D. E. (2016). Patent acquisition, investment, and contracting. *Robert H. Smith School Research Paper No. RHS 2870112*.
- Burchardi, K. B., T. Chaney, T. A. Hassan, L. Tarquinio, and S. J. Terry (2020). Immigration, innovation, and growth. *NBER Working Paper No. 27075*.
- Calomiris, C. W. and U. Khan (2015). An assessment of tarp assistance to financial institution. *Journal of Economic Perspectives* 29(2), 53–80.
- Calvet, L., C. Celerier, P. Sodini, and B. Vallée (2018). Can financial innovation solve household reluctance to take risk? *Harvard Business School Working Paper, No. 18-066*.
- Canayaz, M., I. Erel, U. G. Gurun, and Y. Wu (2024). When protectionism kills talent. *Fisher College of Business Working Paper No. 2024-03-007*.
- Chen, J., S. Hsieh, and F. Zhang (2021). The role of high-skilled foreign labor in startup performance: Evidence from two natural experiments. *Journal of Financial Economics* 142(1), 430–452.
- Chen, M. A., Q. Wu, and B. Yang (2019). How valuable is FinTech innovation? *The Review of Financial Studies* 32(5), 2062–2106.
- Cheng, I.-H., S. Raina, and W. Xiong (2014). Wall Street and the housing bubble. *The American Economic Review* 104(9), 2797–2829.
- Cohn, J. B., Z. Liu, and M. Wardlaw (2022). Count (and count-like) data in finance. *Journal of Financial Economics* 146(2), 529–551.
- Cornaggia, J., K. Cornaggia, and R. D. Israelsen (2024). Unauthorized immigration and local government finances. *Working Paper*.
- Cunningham, C., F. Ederer, and S. Ma (2021). Killer acquisitions. *Journal of Political Economy* 129(3), 649–702.
- D’Acunto, F. (2020). Innovating to invest: The role of basic education. *Working Paper*.
- D’Acunto, F. and L. Frésard (2019). Finance, talent allocation, and growth. *Working Paper*.
- D’Acunto, F., G. Tate, and L. Yang (2021). Correcting market failures in entrepreneurial finance. *Working Paper*.
- D’Acunto, F., G. A. Tate, and L. Yang (2023). Entrepreneurial teams: Diversity of prior industry experience and early-stage growth. *Available at SSRN 3750982*.
- D’Acunto, F., M. Weber, and J. Xie (2019). Punish one teach a hundred: The sobering effect of punishment on the unpunished. *Fama-Miller Working Paper No. 19-06*.
- D’Acunto, F., J. Yao, and J. Xie (2021). Trust and contracts: Empirical evidence. CESifo Working Paper No. 8714.
- Denes, M., S. T. Howell, F. Mezzanotti, X. Wang, and T. Xu (2023). Investor tax credits and entrepreneurship: Evidence from US states. *The Journal of Finance* 78(5), 2621–2671.
- Dimmock, S., J. Huang, and S. Weisbenner (2022). Give me your tired, your poor, your high-skilled labor: H-1B lottery outcomes and entrepreneurial success. *Management Science* 68(9), 6355–7064.

- Frame, W. S. and L. J. White (2004). Empirical studies of financial innovation: Lots of talk, little action? *Journal of Economic Literature* 42(1), 116–144.
- Frame, W. S. and L. J. White (2012). Technological change, financial innovation, and diffusion in banking. *The Oxford Handbook of Banking. Edited by Allen N. Berger, Philip Molyneux, and John O. S. Wilson*, 486–507.
- Genin, A., W. Ma, H. Cao, and X. Song (2022). Intellectual property court and firm innovation in China. In *Academy of Management Proceedings*, Volume 2022, pp. 16517. Academy of Management Briarcliff Manor, NY 10510.
- Glennon, B. (2024a). How do restrictions on high-skilled immigration affect offshoring. *Management Science* 70(2), 907–930.
- Glennon, B. (2024b). Skilled immigrants, firms, and the global geography of innovation. *Journal of Economic Perspectives* 38(1), 3–26.
- Griliches, Z. (2007). *R&D and productivity: The econometric evidence*. University of Chicago Press.
- He, J. and X. Tian (2020). Institutions and innovation: A review of recent literature. *Annual Review of Financial Economics* 12(1), 377–398.
- Hombert, J. and A. Matray (2018). Can innovation help us manufacturing firms escape import competition from China? *The Journal of Finance* 73(5), 2003–2039.
- Hunt, J. (2011). Which immigrants are most innovative and entrepreneurial? Distinctions by entry visa. *Journal of Labor Economics* 29(3), 417–457.
- Hunt, J. and M. Gauthier-Loiselle (2010). How much does immigration boost innovation? *American Economic Journal: Macroeconomics* 2(2), 31–56.
- Jaimovich, N. and H. E. Siu (2017). High-skilled immigration, stem employment, and non-routine-biased technical change. *NBER Working Paper No. 23185*.
- Jeffers, J. S. (2024). The impact of restricting labor mobility on corporate investment and entrepreneurship. *The Review of Financial Studies* 37(1), 1–44.
- Jones, C. I. (1995). R&D-based models of economic growth. *Journal of Political Economy* 103(4), 759–784.
- Kerr, S. P. and W. F. Lincoln (2010). The supply side of innovation: H-1B visa reforms and U.S. ethnic invention. *Journal of Labor Economics* 28(3), 473–508.
- Kerr, W. R. (2008). Ethnic scientific communities and international technology distribution. *Review of Economics and Statistics* 90, 518–537.
- Kogan, L., D. Papanikolaou, A. Seru, and N. Stoffman (2017). Technological innovation, resource allocation, and growth. *The Quarterly Journal of Economics* 132(2), 665 – 712.
- Laeven, L., R. Levine, and S. Michalopoulos (2015). Financial innovation and endogenous growth. *Journal of Financial Intermediation* 24(1), 1–24.
- Lerner, J. (1996). The government as venture capitalist: The long-run effects of the sbir program.
- Lerner, J. (2002). Where does State Street lead? A first look at finance patents, 1971-2000. *The Journal of Finance* 57(2), 901–930.
- Lerner, J. (2006). The new new financial thing: The origins of financial innovations. *Journal of Financial Economics* 79(2), 223–255.
- Lerner, J. (2010). The litigation of financial innovations. *Journal of Law and Economics* 53(4), 807–831.

- Lerner, J., A. Seru, N. Short, and Y. Sun (2023). Financial innovation in the 21st century: Evidence from U.S. patents. *Working paper*.
- Lerner, J. and P. Tufano (2012). The consequences of financial innovation: A counterfactual research agenda. *The Rate and Direction of Inventive Activity Revisited*, 523–575.
- Lucca, D., A. Seru, and F. Trebbi (2014). The revolving door and worker flows in banking regulation. *Journal of Monetary Economics* 65, 17–32.
- Ma, S., J. T. Tong, and W. Wang (2022). Bankrupt innovative firms. *Management Science* 68(9), 6355–7064.
- Mayer, C., E. Morrison, T. Piskorski, and G. Gupta (2014). Mortgage modification and strategic behavior: Evidence from a legal settlement with countrywide. *The American Economic Review* 104(9), 2830–2857.
- Moser, P., A. Voena, and F. Waldinger (2014). German jewish émigrés and US invention. *The American Economic Review* 104, 3222–3255.
- Ouimet, P. and R. Zarutskie (2020). Acquiring labor. *Quarterly Journal of Finance* 10(03), 2050011.
- Peri, G., K. Shih, and C. Sparber (2015). Stem workers, H-1B visas, and productivity in US cities. *Journal of Labor Economics* 33(3), 225–255.
- Pérignon, C. and B. Vallée (2017). The political economy of financial innovation: Evidence from local governments. *The Review of Financial Studies* 30(6), 1903–1934.
- Puri, M., Y. Qian, and X. Zheng (2024). From competitors to partners: Banks’ venture investments in Fintech. *Working Paper*.
- Romer, P. M. (1990). Endogenous technological change. *Journal of Political Economy* 98(5), S71–S102.
- Rysman, M. and S. Schuh (2016). New innovations in payments. *NBER Working Paper No. 22358*.
- Sevilir, M. and X. Tian (2023). Acquiring innovation. *Working paper*.
- Stuen, E. T., A. M. Mobarak, and K. E. Maskus (2012). Skilled immigration and innovation: Evidence from enrolment fluctuations in us doctoral programmes. *The Economic Journal* 122, 1143–1176.
- Tian, X. and J. Xu (2022). Do place-based policies promote local innovation and entrepreneurship? *Review of Finance* 26(3), 595–635.
- Tufano, P. (1989). Financial innovation and first-mover advantages. *Journal of Financial Economics* 25, 213–240.
- Tufano, P. (1995). Securities innovations: A historical and functional perspective. *Journal of Applied Corporate Finance* 7(Winter), 90–103.
- Tufano, P. (2003). Financial innovation. *The Oxford Handbook of Banking*. Edited by Allen N. Berger, Philip Molyneux, and John O. S. Wilson, 307–336.
- Weiss, L. (2014). *America Inc.?: Innovation and enterprise in the national security state*. Cornell University Press.
- Xu, J. (2021). The effect of tax incentives on local private investments and entrepreneurship: Evidence from the Tax Cuts and Jobs Act of 2017. *Available at SSRN 4082335*.

Figure 2: Aggregated Patents and Top 20 Innovative Financial Institutions

This figure presents the number of patents filed by all financial institutions in our sample over time (Panel A), the number of patents filed by the top 20 financial institutions that filed the most patents from January 2007 through December 2014 (Panel B), the number of inventors who filed patents for financial institutions over time (Panel C), and the number of patents filed by the top 20 financial institutions that hired the most inventors from January 2007 through December 2014 (Panel D). In Panels A and C, the x-axes measure time at the year-quarter level; in Panels B and D, the x-axes report institution names. In Panel A, the y-axis is the total number of patents filed in each year-quarter. In Panel B, the y-axis is the total number of patents filed by each of financial institutions. In Panel C, the y-axis is the total number of inventors filing patents for our sample financial institutions in each year-quarter. In Panel D, the y-axis is the total number of inventors filing patents for each of financial institutions.

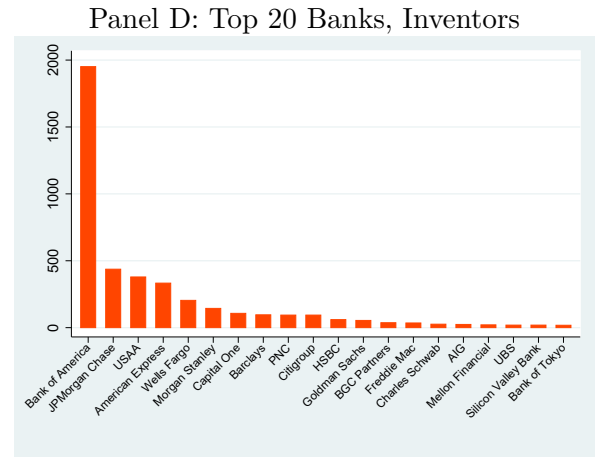
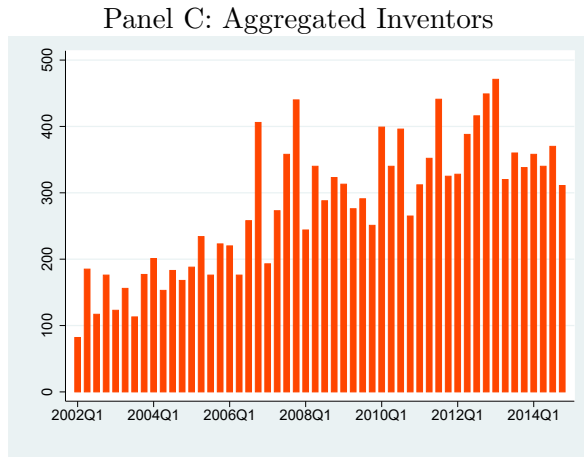
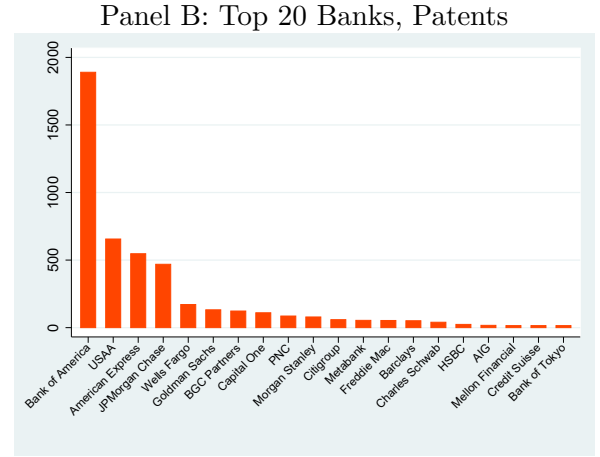
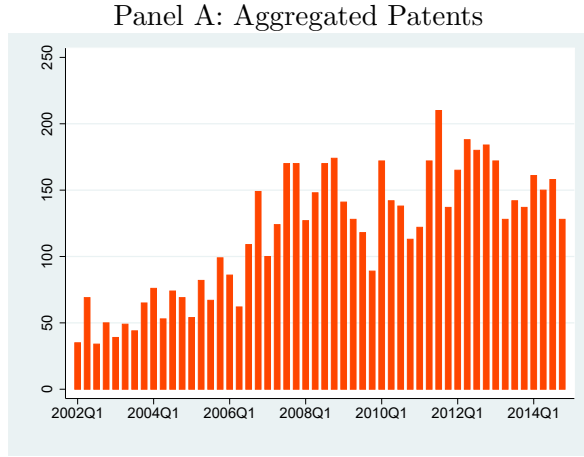


Figure 3: **H1-B STEM Hiring**

The figure plots the average number (Panels A and B) of H-1B STEM jobs that US banks hire and average frequency (Panels C and D) with which US banks hire H-1B STEM jobs (y-axis). The reported averages are calculated across bank-year-month observations within each event quarter (x-axis). Panel A and C include both TARP and non-TARP recipients that sponsored H-1B STEM jobs during 2004-2006. The blue (light-green) line represents TARP (non-TARP) recipients. Panels B and D only include TARP recipients. The blue (light-green) line represents TARP recipients that had high (low) exposure to EAWA (see section 3.1 for how high and low exposures are constructed). The time periods are categorized as “Before,” “EAWA (Event Time),” and “After,” corresponding to quarters before, during, and after banks’ compliance with EAWA. The Event Time period for TARP recipients is institution-specific (refer to Table A.1), whereas the period of non-TARP recipients is calculated from February 17, 2009, through February 17, 2011. See section 2.1 for the selection of our sample banks. Confidence intervals are calculated at the 95% level.

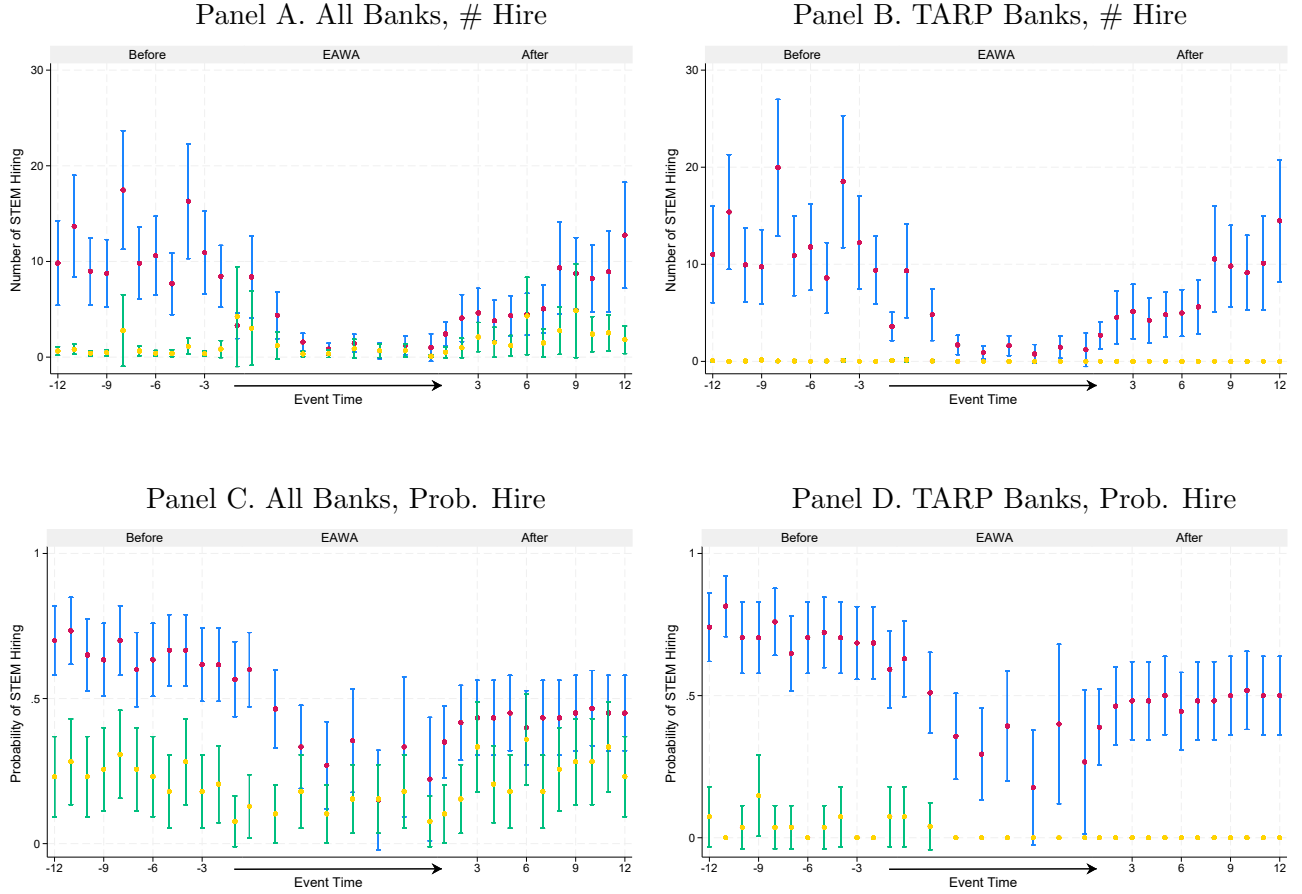


Figure 4: **STEM Skills**

This figure plots the difference in *O*NET* ratings for the level of each skill or knowledge element of STEM occupations relative to non-STEM occupations. We use information from H-1B visas sponsored by our sample financial institutions through the H-1B program over the period of 2004-2006. For each skill or knowledge element h corresponding to institution i sponsoring N STEM occupations and M non-STEM occupations, the “Difference” score is calculated as follows:

$$\text{Difference}_{h,i} = \frac{\frac{\sum_{k=1}^N \sum_{n=1}^n \text{Rating}_{h,k}}{\sum_{k=1}^N \sum_{n=1}^n 1}}{\frac{\sum_{k=1}^M \sum_{m=1}^m \text{Rating}_{h,k}}{\sum_{k=1}^M \sum_{m=1}^m 1}} - 1,$$

where $\text{Rating}_{h,k}$ is the mean of ratings across all survey participants rated by *O*NET* with respect to element h required by occupation k . For each STEM or non-STEM occupation, institution i sponsors n and m individual H-1B visas, respectively. The mean of difference for element h across J banks is calculated as follows:

$$\text{Difference}_h = \frac{\sum_J \text{Difference}_{h,i}}{J}.$$

t -statistics for the mean of each skill or knowledge element h across L financial institutions is provided in the figure. In Panels A and B, the x -axes are skill elements; in Panels C and D, the x -axes are knowledge elements. In Panels A and B, the y -axes are Difference_h ; in Panels C and D, the y -axes are t -statistics.



Table 1: **Qualitative Evidence on Patents Filed by Banks**

Panel A of this table reports the 20 most frequent classifications for patents granted by the US Patent and Trademark Office (USPTO) to the financial institutions in our sample. These patents were filed over the period of 2007-2014. Panel B of this table reports the top 20 patents by citations that were filed by financial institutions in our sample in one year (2012).

Panel A: Top 20 Patent Classification		
Classification	Description	#Patents
705	Data processing: financial, business practice, management, or cost/price determination	2,440
235	Registers	416
709	Electrical computers and digital processing systems: multi-computer data transferring	231
726	Information security	222
707	Data processing: database and file management or data structures	219
382	Image analysis	132
370	Multiplex communications	118
717	Data processing: software development, installation, and management	109
714	Error detection/correction and fault detection/recovery	103
713	Electrical computers and digital processing systems: support	90
398	Optical communications	89
379	Telephonic communications	82
375	Pulse or digital communications	81
715	Data processing: presentation processing of document, operator interface processing, and screen saver display processing	76
455	Telecommunications	75
706	Data processing: artificial intelligence	60
340	Communications: electrical	48
718	Electrical computers and digital processing systems: virtual machine task or process management or task management/control	38
703	Data processing: structural design, modeling, simulation, and emulation	37

Panel B: Top 20 Patents Filed in 2012

Assignee Name	Patent Number	Filing Date	# Citations	Description
JPMorgan Chase	8433652	Mar 29, 2012	705	Method and system for processing internet payments using the electronic funds transfer network
JPMorgan Chase	8781543	Mar 26, 2012	694	Manual and automatic probe calibration
American Express	8401898	Aug 16, 2012	598	System and method for using loyalty rewards as currency
JPMorgan Chase	8602301	Sep 13, 2012	516	Selectable multi-purpose card
Bank of America	8839363	Mar 16, 2012	404	Trusted hardware for attesting to authenticity in a cloud environment
JPMorgan Chase	8469265	Jun 21, 2012	285	Method and system for implementing a card product with multiple customized relationships
American Express	8589255	Nov 6, 2012	279	Virtual reality shopping experience
Bank of America	8666895	Feb 21, 2012	277	Single action mobile transaction device
American Express	8191778	Feb 2, 2012	269	System and method for immediate issuance of transaction cards
American Express	8478639	Oct 23, 2012	248	System and method for a multiple merchant stored value card
JPMorgan Chase	8639017	Sep 14, 2012	157	Method and system for duplicate check detection
Bank of America	9043609	Jul 19, 2012	152	Implementing security measures for authorized tokens used in mobile transactions
Bank of America	8919643	Sep 14, 2012	146	Method and apparatus for using at least a portion of a one-time password as a dynamic card verification value
JPMorgan Chase	8826371	Aug 6, 2012	139	Authentication system and method
American Express	8572712	Mar 7, 2012	129	Device independent authentication system and method
American Express	8401889	Mar 28, 2012	124	Estimating the spend capacity of consumer households
American Express	8401539	Jan 31, 2012	118	Servicing attributes on a mobile device
JPMorgan Chase	8548886	Apr 26, 2012	113	Account opening system, method and computer program product
BGC Partners	8732069	Sep 14, 2012	111	Systems and methods for monitoring credit of trading counterparties

Table 2: Sample Descriptive Statistics

This table presents descriptive statistics for our working samples. The unit is bank i as of year-month s . The sample period is 2007-2014. We require banks to have submitted at least one patent application with the US patent and trademark office (USPTO) before December 31, 2006. Panel A describes the sample for our first empirical strategy, which includes 20 TARP and 13 non-TARP banks, all of which had hired at least one H-1B STEM worker during 2004-2006. Panel B describes the sample for our second empirical strategy, which includes 27 TARP banks. TARP is a dummy variable that equals 1 if bank i accessed the Troubled Asset Relief Program (TARP), and 0 otherwise. EAWA is a dummy variable that equals 1 if bank i was subject to the Employ American Workers Act (EAWA) in year-month s , and 0 otherwise. Post is a dummy variable that equals 1 for all year-months after the EAWA ban ended, and 0 otherwise. Patent Filed is a dummy variable that equals 1 if bank i filed at least one granted patent in year-month s , and 0 otherwise. Patent Purchase is a dummy variable that equals 1 if bank i engaged with at least one inter-firm patent purchase in year-month s , and 0 otherwise. H1B STEM is a dummy variable that equals 1 if bank i hired at least one H-1B-sponsored STEM worker in year-month s , and 0 otherwise. #H1B STEM is the number of H-1B-sponsored STEM workers hired by the bank i in year-month s . First Filer is a dummy variable that equals 1 if patents filed by bank i in year-month s include at least one investor who filed for the first time for bank i , and 0 otherwise. Non-First Filer is a dummy variable that equals 1 if all patents filed by bank i in year-month s do not have any investors who filed for the first time for bank i , and 0 otherwise. First Filer Lead is a dummy variable that equals 1 if patents filed by bank i in year-month s include at least one inventor who not only filed for the first time for bank i but also was listed as the lead inventor, and 0 otherwise. H1B₋₃ is a dummy variable that equals 1 if bank i as of year-month s hired at least one H-1B-sponsored worker over the previous three years, and 0 otherwise. STEM₋₃ is a dummy variable that equals 1 if bank i as of year-month s hired at least one H-1B-sponsored STEM worker over the previous three years, and 0 otherwise. TARP Fund% is the amount of funding over total assets measured in 2009 that TARP bank i obtained via the program. High Exposure _{i} is a dummy variable that equals 1 if the number of H-1B STEM workers bank i hired in 2004-2006 is above the median of the distribution over a subsample of banks of similar size. Banks are classified into small, medium, and large if their assets are below \$10 billion, between US\$10 and US\$50 billion, and above US\$1 trillion, respectively.

	N	Mean	Std	Min	P10	P25	P50	P75	P90	Max
Panel A. All Banks										
TARP	3,168	0.606	0.489	0	0	0	1	1	1	1
EAWA	3,168	0.094	0.292	0	0	0	0	0	0	1
Post	3,168	0.349	0.477	0	0	0	0	1	1	1
Patent Filed	3,168	0.205	0.404	0	0	0	0	0	1	1
Patent Purchase	3,168	0.048	0.214	0	0	0	0	0	0	1
H1B STEM	3,168	0.376	0.484	0	0	0	0	1	1	1
#H1B STEM	3,168	6.874	23.440	0	0	0	0	4	19	392
First Filer	3,168	0.148	0.355	0	0	0	0	0	1	1
First Filer Lead	3,168	0.085	0.279	0	0	0	0	0	0	1
Non First Filer	3,168	0.058	0.233	0	0	0	0	0	0	1
H1B ₋₃	3,168	0.840	0.367	0	0	1	1	1	1	1
STEM ₋₃	3,168	0.753	0.431	0	0	1	1	1	1	1
Panel B. TARP Banks										
High Exposure	2,592	0.667	0.471	0	0	0	1	1	1	1
TARP Fund%	2,592	0.025	0.023	0	0.011	0.013	0.021	0.026	0.063	0.110
EAWA	2,592	0.159	0.365	0	0	0	0	0	1	1
Post	2,592	0.571	0.495	0	0	0	1	1	1	1
Patent Filed	2,592	0.247	0.432	0	0	0	0	0	1	1
Patent Purchase	2,592	0.061	0.239	0	0	0	0	0	0	1
H1B STEM	2,592	0.361	0.480	0	0	0	0	1	1	1
#H1B STEM	2,592	7.428	24.530	0	0	0	0	4	21	392
First Filer	2,592	0.177	0.382	0	0	0	0	0	1	1
First Filer Leader	2,592	0.102	0.303	0	0	0	0	0	1	1
Non First Filer	2,592	0.070	0.255	0	0	0	0	0	0	1
H1B ₋₃	2,592	0.678	0.467	0	0	0	1	1	1	1
STEM ₋₃	2,592	0.633	0.482	0	0	0	1	1	1	1

Table 3: **Employ American Workers Act (EAWA) and Hiring of Foreign STEM Workers**

This table reports the estimates of the effect of EAWA on foreign STEM hiring. Columns (1)-(4) report estimates from the following ordinary least squares (OLS) specification:

$$Y_{i,s} = \alpha + \beta_1 EAWA_{i,s} + \beta_2 EAWA_{i,s} \times TARP_i + \beta_3 Post_{i,s} + \beta_4 Post_{i,s} \times TARP_i + X'_i \times \theta + \eta_i + \eta_s + \epsilon_{i,s}.$$

The sample in columns (1)-(4) consists of 20 TARP and 13 non-TARP banks that sponsored at least one STEM job through the H-1B program during 2004-2006. Columns (5)-(8) report estimates from the following OLS specification:

$$Y_{i,s} = \alpha + \beta_1 EAWA_{i,s} + \beta_2 EAWA_{i,s} \times High\ Exposure_i + \beta_3 Post_{i,s} + \beta_4 Post_{i,s} \times High\ Exposure_i + X'_i \times \theta + \eta_i + \eta_s + \epsilon_{i,s}.$$

The sample in columns (5)-(8) consists of 27 TARP banks. In Panel A, the dependent variable is $\ln(\#H1B\ STEM + 0.001)_{i,s}$, which is the logarithm of the number of H-1B STEM workers sponsored by bank i in year-month s plus 0.001. In Panel B, the dependent variable is a dummy variable ($H-1B\ STEM_{i,s}$) that equals 1 if bank i sponsored at least one H-1B visa in year-month s , and 0 otherwise. See Table 2 for definitions of other variables. Standard errors are clustered at the bank level.

Panel A: Dependent Variable: $\ln(\#H1B\ STEM + 0.001)$

	All Banks				TARP Banks			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
EAWA $\times$ TARP	-2.954*** (-3.87)	-3.202*** (-2.98)	-2.455*** (-3.89)	-1.828** (-2.19)				
Post $\times$ TARP	-0.548 (-0.61)	-2.052** (-2.09)	-1.940** (-2.38)	-2.302** (-2.66)				
EAWA $\times$ High Exposure					-3.566*** (-3.93)	-3.671*** (-3.99)	-2.502*** (-3.44)	-2.605*** (-3.62)
EAWA					0.960 (1.19)	3.040*** (2.82)	-0.058 (-0.09)	1.590** (2.40)
Post $\times$ High Exposure					-1.613 (-1.53)	-1.532 (-1.40)	-1.889** (-2.07)	-1.842* (-1.98)
Post					0.317 (0.47)	1.393 (0.78)	-0.076 (-0.25)	0.759 (1.40)
EAWA $\times$ TARP Fund%					-29.974 (-1.48)	-32.121 (-1.63)	-4.423 (-0.26)	-6.538 (-0.42)
Post $\times$ TARP Fund%					-1.613 (-1.53)	-1.532 (-1.40)	-1.889** (-2.07)	-1.842* (-1.98)
TARP	3.642*** (2.89)	4.515*** (3.50)						
High Exposure					4.909*** (3.66)	4.617*** (3.43)		
TARP Fund%					12.483 (0.46)	10.892 (0.38)		
H1B ₋₃	1.402 (1.65)	1.600* (1.79)	0.340 (0.99)	0.639 (1.45)	0.862 (1.38)	0.573 (0.88)	0.289 (0.91)	0.234 (0.78)
STEM ₋₃	4.285*** (4.66)	4.619*** (4.63)	0.547** (2.58)	0.778** (2.23)	3.807*** (3.47)	4.457*** (3.86)	0.490 (1.22)	0.829 (1.62)
Constant	-11.139*** (-12.68)	-11.538*** (-12.56)	-4.787*** (-9.66)	-5.145*** (-9.35)	-11.072*** (-11.67)	-12.038*** (-7.78)	-4.471*** (-8.78)	-5.403*** (-9.83)
Year $\times$ Month FE	No	Yes	No	Yes	No	Yes	No	Yes
Bank FE	No	No	Yes	Yes	No	No	Yes	Yes
N	3,168	3,168	3,168	3,168	2,592	2,592	2,592	2,592
adj. R ²	0.27	0.30	0.66	0.69	0.40	0.41	0.74	0.76

Panel B: Dependent Variable: Probability of H-1B STEM Hiring

	All Banks				TARP Banks			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
EAWA $\times$ TARP	-0.239*** (-3.32)	-0.261** (-2.57)	-0.198*** (-3.33)	-0.144* (-1.81)				
Post $\times$ TARP	-0.059 (-0.72)	-0.182* (-2.03)	-0.181** (-2.37)	-0.202** (-2.50)				
EAWA $\times$ High Exposure					-0.289*** (-3.46)	-0.299*** (-3.55)	-0.199*** (-2.84)	-0.208*** (-3.02)
EAWA					0.087 (1.11)	0.252** (2.48)	0.001 (0.02)	0.127* (1.84)
Post $\times$ High Exposure					-0.140 (-1.48)	-0.135 (-1.39)	-0.165* (-1.90)	-0.163* (-1.85)
Post					0.009 (0.16)	0.121 (0.80)	-0.027 (-0.76)	0.072 (1.32)
EAWA $\times$ TARP Fund%					-2.770 (-1.36)	-2.932 (-1.49)	-0.688 (-0.40)	-0.844 (-0.53)
Post $\times$ TARP Fund%					1.297 (1.30)	1.435 (1.41)	-0.076 (-0.06)	-0.012 (-0.01)
TARP	0.306*** (2.84)	0.377*** (3.37)						
High Exposure					0.420*** (3.72)	0.402*** (3.54)		
TARP Fund%					0.655 (0.35)	0.557 (0.28)		
H1B ₋₃	0.115 (1.65)	0.127* (1.76)	0.031 (0.97)	0.046 (1.19)	0.077 (1.48)	0.057 (1.07)	0.020 (0.64)	0.017 (0.57)
STEM ₋₃	0.387*** (4.85)	0.415*** (4.90)	0.051** (2.52)	0.068** (2.16)	0.332*** (3.53)	0.374*** (3.84)	0.052 (1.29)	0.064 (1.30)
Constant	-0.154** (-2.10)	-0.184** (-2.44)	0.393*** (8.55)	0.370*** (7.34)	-0.141* (-2.01)	-0.232* (-1.89)	0.417*** (8.77)	0.336*** (6.61)
Year $\times$ Month FE	No	Yes	No	Yes	No	Yes	No	Yes
Bank FE	No	No	Yes	Yes	No	No	Yes	Yes
N	3,168	3,168	3,168	3,168	2,592	2,592	2,592	2,592
adj. R ²	0.26	0.28	0.61	0.63	0.40	0.40	0.70	0.71

Table 4: **Employ American Workers Act (EAWA) and Patenting Activities**

This table reports the estimates of the effect of EAWA on patenting activities. Columns (1)-(4) report estimates from the following ordinary least squares (OLS) specification:

$$\text{Patent Filed}_{i,s} = \alpha + \beta_1 \text{EAWA}_{i,s} + \beta_2 \text{EAWA}_{i,s} \times \text{TARP}_i + \beta_3 \text{Post}_{i,s} + \beta_4 \text{Post}_{i,s} \times \text{TARP}_i + X'_i \times \theta + \eta_i + \eta_s + \epsilon_{i,s}.$$

The sample in columns (1)-(4) consists of 20 TARP and 13 non-TARP banks that sponsored at least one STEM job through the H-1B program during 2004-2006. Columns (5)-(8) report estimates from the following OLS specification:

$$\text{Patent Filed}_{i,s} = \alpha + \beta_1 \text{EAWA}_{i,s} + \beta_2 \text{EAWA}_{i,s} \times \text{High Exposure}_i + \beta_3 \text{Post}_{i,s} + \beta_4 \text{Post}_{i,s} \times \text{High Exposure}_i + X'_i \times \theta + \eta_i + \eta_s + \epsilon_{i,s}.$$

The sample in columns (5)-(8) consists of 27 TARP banks. $\text{Patent Filed}_{i,s}$ is a dummy variable that equals 1 if bank i filed at least one granted patent in year-month s , and 0 otherwise. See Table 2 for definitions of other variables. Standard errors are clustered at the bank level.

	All Banks				TARP Banks			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
EAWA $\times$ TARP	-0.123*** (-3.81)	-0.177*** (-3.33)	-0.065** (-2.50)	-0.059* (-1.84)				
Post $\times$ TARP	0.075* (1.80)	0.049 (1.05)	-0.020 (-0.79)	0.003 (0.10)				
EAWA $\times$ High Exposure					-0.175*** (-3.02)	-0.176*** (-2.94)	-0.103** (-2.50)	-0.101** (-2.47)
EAWA					0.065 (1.12)	0.125 (1.17)	0.032 (0.80)	-0.010 (-0.19)
Post $\times$ High Exposure					-0.034 (-0.61)	-0.032 (-0.56)	-0.053* (-1.91)	-0.054* (-1.95)
Post					0.053 (0.96)	0.200 (1.48)	0.016 (0.79)	0.007 (0.20)
EAWA $\times$ TARP Fund%					-1.380 (-1.40)	-1.265 (-1.20)	-0.173 (-0.31)	-0.108 (-0.19)
Post $\times$ TARP Fund%					1.322 (1.36)	1.266 (1.25)	0.257 (0.55)	0.224 (0.51)
TARP	0.210** (2.48)	0.234** (2.52)						
High Exposure					0.162 (1.65)	0.160 (1.58)		
TARP Fund%					0.439 (0.21)	0.478 (0.22)		
H1B ₋₃	0.106 (1.66)	0.113 (1.66)	-0.007 (-0.46)	-0.005 (-0.23)	0.009 (0.12)	0.001 (0.01)	-0.009 (-0.54)	-0.010 (-0.69)
STEM ₋₃	0.166*** (3.03)	0.168*** (2.94)	-0.006 (-0.71)	-0.017 (-1.25)	0.232** (2.71)	0.237** (2.57)	-0.004 (-0.25)	-0.012 (-0.54)
Constant	-0.151** (-2.20)	-0.159** (-2.21)	0.229*** (10.82)	0.227*** (8.78)	-0.048 (-0.53)	-0.139 (-1.17)	0.270*** (12.81)	0.288*** (12.24)
Year $\times$ Month FE	No	Yes	No	Yes	No	Yes	No	Yes
Bank FE	No	No	Yes	Yes	No	No	Yes	Yes
N	3,168	3,168	3,168	3,168	2,592	2,592	2,592	2,592
adj. R ²	0.16	0.15	0.62	0.61	0.13	0.12	0.60	0.59

Table 5: **Employ American Workers Act (EAWA) and Patenting Activities by Patent Categories**

This table reports the estimates of the effect of EAWA on patenting activities. Columns (1)-(3) report estimates from the following ordinary least squares (OLS) specification:

$$\text{Patent Filed}_{i,s} = \alpha + \beta_1 \text{EAWA}_{i,s} + \beta_2 \text{EAWA}_{i,s} \times \text{TARP}_i + \beta_3 \text{Post}_{i,s} + \beta_4 \text{Post}_{i,s} \times \text{TARP}_i + X'_i \times \theta + \eta_i + \eta_s + \epsilon_{i,s}.$$

The sample in columns (1)-(3) consists of 20 TARP and 13 non-TARP banks that sponsored at least one STEM job through the H-1B program during 2004-2006. Columns (4)-(6) report estimates from the following OLS specification:

$$\text{Patent Filed}_{i,s} = \alpha + \beta_1 \text{EAWA}_{i,s} + \beta_2 \text{EAWA}_{i,s} \times \text{High Exposure}_i + \beta_3 \text{Post}_{i,s} + \beta_4 \text{Post}_{i,s} \times \text{High Exposure}_i + X'_i \times \theta + \eta_i + \eta_s + \epsilon_{i,s}.$$

The sample in columns (4)-(6) consists of 27 TARP banks. $\text{Patent-Filed}_{i,s}$ is a dummy variable that equals 1 if bank i filed at least one granted patent in year-month s , and 0 otherwise. For each patent category (i.e., business method (705), non-business method (Non-705), and FinTech), $\text{Patent Filed}_{i,s}$ is a dummy variable that equals 1 if bank i filed at least one granted patent in that category in year-month s , and 0 otherwise. See Table 2 for definitions of other variables. Standard errors are clustered at the bank level.

	All Banks			TARP Banks		
	705	Non 705	FinTech	705	Non 705	FinTech
	(1)	(2)	(3)	(4)	(5)	(6)
EAWA $\times$ TARP	-0.061 (-1.57)	-0.010 (-0.37)	-0.014 (-0.62)			
Post $\times$ TARP	-0.068 (-1.65)	0.062** (2.52)	0.044** (2.05)			
EAWA $\times$ High Exposure				-0.076* (-1.93)	-0.057* (-1.79)	-0.045* (-1.81)
EAWA				0.000 (0.01)	-0.008 (-0.14)	-0.002 (-0.03)
Post $\times$ High Exposure				-0.139*** (-3.35)	0.055* (1.88)	0.051* (1.95)
Post				0.126** (2.54)	-0.077 (-1.45)	-0.073 (-1.28)
EAWA $\times$ TARP Fund%				-0.289 (-0.84)	-0.092 (-0.22)	-0.643 (-1.13)
Post $\times$ TARP Fund%				-0.847* (-1.81)	0.693 (1.50)	-0.559 (-1.38)
H1B ₋₃	-0.093** (-2.65)	0.042** (2.09)	0.039** (2.09)	-0.004 (-0.13)	-0.007 (-0.40)	0.017 (0.79)
STEM ₋₃	-0.036 (-1.40)	0.014 (0.84)	0.020 (1.45)	-0.104*** (-3.20)	0.043* (1.73)	0.032 (1.24)
Constant	0.263*** (6.84)	0.089*** (3.32)	0.049* (1.81)	0.228*** (6.19)	0.185*** (6.44)	0.140*** (3.89)
Year $\times$ Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes
N	3,168	3,168	3,168	2,592	2,592	2,592
adj. R ²	0.48	0.57	0.51	0.47	0.56	0.50

Table 6: **Employ American Workers Act (EAWA) and Patent Citations**

This table reports the estimates of the effect of EAWA on patent citations. Columns (1)-(4) report estimates from the following ordinary least squares (OLS) specification:

$$\overline{Adj.Cites}_{i,s} = \alpha + \beta_1 EAWA_{i,s} + \beta_2 EAWA_{i,s} \times TARP_i + \beta_3 Post_{i,s} + \beta_4 Post_{i,s} \times TARP_i + X'_i \times \theta + \eta_i + \eta_s + \epsilon_{i,s}.$$

The sample in columns (1)-(4) consists of 20 TARP and 17 non-TARP banks that sponsored at least one STEM job through the H-1B program during 2004-2006. Columns (5)-(8) report estimates from the following OLS specification:

$$\overline{Adj.Cites}_{i,s} = \alpha + \beta_1 EAWA_{i,s} + \beta_2 EAWA_{i,s} \times High\ Exposure_i + \beta_3 Post_{i,s} + \beta_4 Post_{i,s} \times High\ Exposure_i + X'_i \times \theta + \eta_i + \eta_s + \epsilon_{i,s}.$$

The sample in columns (5)-(8) consists of 27 TARP banks. For each patent category (i.e., all patents, business method (705), non-business method (Non-705), and FinTech), $\overline{Adj.Cites}$ is the averaged $Adj.Cites$ for patents filed by bank i in year-month s . $\overline{Adj.Cites}$ is transformed into percentile ranks. $Adj.Cites$ is the number of citation counts divided by the average number of citation counts in a given technology-class-year (i.e., the year in which all patents of the same technology were applied). Technology domain is defined at the 4-digit International Patent Classification. The end of the observation period for patent citation is January, 2022. See Table A.3 for definitions of other variables. Standard errors are clustered at the bank level.

	All Banks				TARP Banks			
	All	705	Non 705	FinTech	All	705	Non 705	FinTech
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
EAWA $\times$ TARP	-0.022 (-0.51)	0.069 (0.73)	-0.058 (-0.76)	-0.070 (-0.41)				
Post $\times$ TARP	0.005 (0.08)	-0.017 (-0.29)	-0.004 (-0.09)	0.108 (0.95)				
EAWA $\times$ High Exposure					0.085** (2.51)	0.268* (1.92)	-0.217** (-2.89)	-0.555** (-2.63)
EAWA					-0.028 (-0.44)	-0.240*** (-2.98)	0.264*** (3.44)	0.380 (1.46)
Post $\times$ High Exposure					0.042 (0.83)	0.025 (0.38)	-0.065 (-1.37)	-0.594*** (-12.56)
Post					-0.008 (-0.10)	-0.086 (-0.95)	0.119* (1.86)	0.601*** (14.15)
EAWA $\times$ TARP Fund%					-0.854 (-1.06)	0.523 (0.95)	-4.490** (-2.83)	2.924 (0.35)
Post $\times$ TARP Fund%					1.442*** (3.47)	0.267 (0.65)	0.231 (0.56)	2.620*** (3.37)
H1B ₋₃	-0.053 (-0.84)	-0.340*** (-2.96)	-0.279* (-2.03)	-0.732*** (-4.92)	-0.203*** (-2.86)	-0.335*** (-3.85)	-0.265* (-1.85)	-0.732*** (-4.82)
STEM ₋₃	0.034 (0.45)	0.245** (2.65)			0.161** (2.56)	0.151 (1.56)		
Constant	0.495*** (7.90)	0.597*** (10.18)	0.788*** (5.93)	1.197*** (6.65)	0.473*** (7.88)	0.716*** (8.96)	0.699*** (6.19)	1.171*** (6.00)
Year $\times$ Month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	651	404	493	354	641	404	484	348
adj. R ²	0.19	0.35	0.15	0.16	0.20	0.32	0.18	0.17

Table 7: **Employ American Workers Act (EAWA) and Patents' First Filers**

This table reports the estimates of the effect of EAWA on the likelihood of patents being filed by first-time filers. Columns (1)-(4) report estimates from the following ordinary least squares (OLS) specification:

$$\text{First Filer}_{i,s} = \alpha + \beta_1 \text{EAWA}_{i,s} + \beta_2 \text{EAWA}_{i,s} \times \text{TARP}_i + \beta_3 \text{Post}_{i,s} + \beta_4 \text{Post}_{i,s} \times \text{TARP}_i + X'_i \times \theta + \eta_i + \eta_s + \epsilon_{i,s}.$$

The sample in columns (1)-(4) consists of 20 TARP and 13 non-TARP banks that sponsored at least one STEM job through the H-1B program during 2004-2006. Columns (5)-(8) report estimates from the following OLS specification:

$$\text{First Filer}_{i,s} = \alpha + \beta_1 \text{EAWA}_{i,s} + \beta_2 \text{EAWA}_{i,s} \times \text{High Exposure}_i + \beta_3 \text{Post}_{i,s} + \beta_4 \text{Post}_{i,s} \times \text{High Exposure}_i + X'_i \times \theta + \eta_i + \eta_s + \epsilon_{i,s}.$$

The sample in columns (5)-(8) consists of 27 TARP banks. In Panel A, the dependent variable is $\text{First Filer}_{i,s}$, which is a dummy variable that equals 1 if patents filed by bank i in year-month s include at least one investor who filed for the first time for bank i , and 0 otherwise. In Panel B, the dependent variable is $\text{Non-First Filer}_{i,s}$, which is a dummy variable that equals 1 if all patents filed by bank i in year-month s do not have any investors who filed for the first time for bank i , and 0 otherwise. See Table 2 for definitions of other variables. Standard errors are clustered at the bank level.

Panel A: Dependent Variable: First Filer

	All Banks				TARP Banks			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
EAWA $\times$ TARP	-0.111*** (-3.57)	-0.124*** (-2.83)	-0.074*** (-3.35)	-0.046 (-1.68)				
Post $\times$ TARP	0.017 (0.49)	-0.002 (-0.06)	-0.050* (-1.80)	-0.033 (-1.07)				
EAWA $\times$ High Exposure					-0.145*** (-2.95)	-0.149*** (-2.92)	-0.093** (-2.70)	-0.094** (-2.76)
EAWA					0.022 (0.48)	0.078 (0.86)	0.010 (0.29)	0.013 (0.22)
Post $\times$ High Exposure					-0.034 (-0.61)	-0.031 (-0.55)	-0.055 (-1.53)	-0.054 (-1.48)
Post					0.043 (0.77)	0.109 (0.97)	0.010 (0.33)	-0.006 (-0.12)
EAWA $\times$ TARP Fund%					-0.159 (-0.19)	-0.127 (-0.15)	0.142 (0.26)	0.138 (0.23)
Post $\times$ TARP Fund%					-0.362 (-0.36)	-0.341 (-0.33)	-0.848** (-2.75)	-0.832** (-2.74)
TARP	0.181** (2.50)	0.195** (2.39)						
High Exposure					0.127 (1.61)	0.119 (1.49)		
TARP Fund%					-0.821 (-0.60)	-0.851 (-0.60)		
H1B ₋₃	0.081* (1.74)	0.095* (1.87)	-0.003 (-0.17)	0.013 (0.72)	0.057 (1.17)	0.048 (0.98)	0.023 (1.37)	0.024 (1.33)
STEM ₋₃	0.113*** (2.77)	0.112** (2.61)	-0.013 (-1.16)	-0.022 (-1.67)	0.122* (1.78)	0.139* (1.89)	-0.054*** (-3.06)	-0.049* (-2.01)
Constant	-0.110** (-2.06)	-0.122** (-2.14)	0.184*** (8.34)	0.169*** (7.18)	0.004 (0.06)	-0.043 (-0.47)	0.231*** (10.31)	0.236*** (6.88)
Year $\times$ Month FE	No	Yes	No	Yes	No	Yes	No	Yes
Bank FE	No	No	Yes	Yes	No	No	Yes	Yes
N	3,168	3,168	3,168	3,168	2,592	2,592	2,592	2,592
adj. R ²	0.11	0.10	0.51	0.51	0.09	0.07	0.49	0.48

Panel B: Dependent Variable: Non-First Filer

	All Banks				TARP Banks			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
EAWA $\times$ TARP	-0.012 (-0.83)	-0.053* (-1.94)	0.009 (0.56)	-0.013 (-0.65)				
Post $\times$ TARP	0.058** (2.61)	0.052** (2.34)	0.030* (1.94)	0.036** (2.27)				
EAWA $\times$ High Exposure					-0.029 (-1.37)	-0.027 (-1.25)	-0.010 (-0.43)	-0.007 (-0.29)
EAWA					0.043* (1.96)	0.047 (1.53)	0.023 (1.01)	-0.023 (-1.14)
Post $\times$ High Exposure					0.001 (0.03)	-0.000 (-0.01)	0.002 (0.09)	-0.000 (-0.01)
Post					0.010 (0.62)	0.091** (2.08)	0.006 (0.29)	0.014 (0.43)
EAWA $\times$ TARP Fund%					-1.221** (-2.22)	-1.138* (-2.05)	-0.315 (-0.57)	-0.245 (-0.48)
Post $\times$ TARP Fund%					1.684*** (7.09)	1.607*** (6.39)	1.104** (2.44)	1.057** (2.44)
TARP	0.029 (1.22)	0.039 (1.53)						
High Exposure					0.035 (1.25)	0.041 (1.34)		
TARP Fund%					1.260 (1.29)	1.329 (1.38)		
H1B ₋₃	0.025 (1.08)	0.019 (0.79)	-0.004 (-0.46)	-0.018 (-1.29)	-0.048 (-1.08)	-0.047 (-1.10)	-0.032*** (-3.08)	-0.034** (-2.73)
STEM ₋₃	0.054** (2.48)	0.056** (2.54)	0.006 (1.36)	0.005 (0.84)	0.110*** (3.00)	0.098** (2.52)	0.050*** (3.57)	0.037** (2.32)
Constant	-0.041 (-1.62)	-0.037 (-1.48)	0.045*** (3.30)	0.057*** (3.95)	-0.052 (-1.62)	-0.097** (-2.22)	0.039** (2.68)	0.052** (2.20)
Year $\times$ Month FE	No	Yes	No	Yes	No	Yes	No	Yes
Bank FE	No	No	Yes	Yes	No	No	Yes	Yes
N	3,168	3,168	3,168	3,168	2,592	2,592	2,592	2,592
adj. R ²	0.04	0.04	0.19	0.19	0.07	0.07	0.18	0.18

Table 8: **Employ American Workers Act (EAWA) and External Patent Purchases**

This table reports the estimates of the effect of EAWA on patent purchase. Columns (1)-(4) report estimates from the following ordinary least squares (OLS) specification:

$$\text{Patent Purchased}_{i,s} = \alpha + \beta_1 \text{EAWA}_{i,s} + \beta_2 \text{EAWA}_{i,s} \times \text{TARP}_i + \beta_3 \text{Post}_{i,s} + \beta_4 \text{Post}_{i,s} \times \text{TARP}_i + X'_i \times \theta + \eta_i + \eta_s + \epsilon_{i,s}.$$

The sample in columns (1)-(4) consists of 20 TARP and 13 non-TARP banks that sponsored at least one STEM job through the H-1B program during 2004-2006. Columns (5)-(8) report estimates from the following OLS specification:

$$\text{Patent Purchased}_{i,s} = \alpha + \beta_1 \text{EAWA}_{i,s} + \beta_2 \text{EAWA}_{i,s} \times \text{High Exposure}_i + \beta_3 \text{Post}_{i,s} + \beta_4 \text{Post}_{i,s} \times \text{High Exposure}_i + X'_i \times \theta + \eta_i + \eta_s + \epsilon_{i,s}.$$

The sample in columns (5)-(8) consists of 27 TARP banks. $\text{Patent Purchased}_{i,s}$ is a dummy variable that equals to 1 if bank i purchased at least one granted patent in year-month s , and 0 otherwise. See Table 2 for definitions of other variables. Standard errors are clustered at the bank level.

	All Banks				TARP Banks			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
EAWA $\times$ TARP	-0.002 (-0.07)	-0.022 (-0.91)	0.011 (0.49)	0.000 (0.01)				
Post $\times$ TARP	0.023 (0.64)	0.017 (0.56)	0.006 (0.18)	0.006 (0.20)				
EAWA $\times$ High Exposure					-0.034 (-1.28)	-0.038 (-1.33)	-0.013 (-0.51)	-0.016 (-0.60)
EAWA					0.021 (0.90)	0.028 (0.71)	0.019 (0.88)	0.018 (0.56)
Post $\times$ High Exposure					0.000 (0.00)	-0.000 (-0.01)	-0.009 (-0.27)	-0.010 (-0.28)
Post					0.028 (1.06)	0.058 (1.20)	0.019 (0.89)	0.033 (1.46)
EAWA $\times$ TARP Fund%					0.123 (0.21)	0.159 (0.23)	0.051 (0.08)	0.068 (0.10)
Post $\times$ TARP Fund %					-0.057 (-0.10)	-0.056 (-0.10)	-0.092 (-0.19)	-0.089 (-0.18)
TARP	0.062** (2.46)	0.070** (2.55)						
High Exposure					0.030 (1.13)	0.030 (1.08)		
TARP Fund %					-0.278 (-0.57)	-0.285 (-0.55)		
H1B ₋₃	0.040* (1.81)	0.046* (1.79)	0.013 (1.11)	0.020 (1.21)	0.029 (1.25)	0.028 (1.13)	0.014 (0.84)	0.016 (0.79)
STEM ₋₃	0.019 (1.28)	0.017 (1.08)	-0.000 (-0.03)	-0.004 (-0.39)	0.042 (1.30)	0.043 (1.18)	0.000 (0.02)	-0.001 (-0.07)
Constant	-0.046 (-1.64)	-0.049 (-1.58)	0.034 (1.28)	0.032 (1.14)	-0.014 (-0.76)	-0.032 (-0.96)	0.044 (1.61)	0.036* (1.87)
Year $\times$ Month FE	No	Yes	No	Yes	No	Yes	No	Yes
Bank FE	No	No	Yes	Yes	No	No	Yes	Yes
N	3,168	3,168	3,168	3,168	2,592	2,592	2,592	2,592
adj. R ²	0.04	0.03	0.31	0.31	0.03	0.02	0.28	0.28

Table 9: **Employ American Workers Act (EAWA) and Wage Premia of Foreign High-Skill Workers**

This table reports estimates from the following ordinary least squares (OLS) specification:

$$\begin{aligned} \overline{Wage\ Premia}_{i,l,s} = & \alpha + \beta_1 EAWA_{i,s} \times TARP_i \times STEM_{i,l,s} + \beta_2 EAWA_{i,s} \times TARP_i + \beta_3 EAWA_{i,s} \times STEM_{i,l,s} \\ & + \beta_4 EAWA_{i,s} + \beta_5 Post_{i,s} \times TARP_i \times STEM_{i,l,s} + \beta_6 Post_{i,s} \times TARP_i + \beta_7 Post_{i,s} \times STEM_{i,l,s} + \\ & \beta_8 Post_{i,s} + \beta_9 STEM_{i,l,s} \times TARP_i + \beta_{10} STEM_{i,l,s} + \beta_{11} TARP_i + \eta_l + \eta_i + \eta_s + \epsilon_{i,l,s}. \end{aligned}$$

The sample consists of TARP and non-TARP banks that sponsored at least one visa for STEM jobs through the H-1B program from January 2004 through December 2006. $\overline{Wage\ Premia}_{i,l,s}$ is the weighted-average wage premia for H-1B visas within the same job category l , submitted by bank i in year-month s . The weight is determined by the number of immigrants sponsored under each H-1B visa. Wage premium of each visa is wage divided by prevailing wage. $STEM_{i,l,s}$ is a dummy variable that equals 1 if job category l for visas sponsored by bank i in year-month s falls into the STEM classification, and 0 otherwise. See Table A.3 for definitions of other variables. Standard errors are clustered at the bank level.

Panel A: Descriptive Statistics

	N	Mean	Std	Min	P10	P25	P50	P75	P90	Max
Wage Premia	8,981	1.266	0.208	1	1.068	1.122	1.211	1.348	1.541	2.091
EAWA	8,981	0.088	0.283	0	0	0	0	0	0	1
Post	8,981	0.540	0.498	0	0	0	1	1	1	1
STEM Job	8,981	0.511	0.500	0	0	0	1	1	1	1
TARP	8,981	0.900	0.299	0	1	1	1	1	1	1

Panel B: Regression Analysis

	(1)	(2)	(3)	(4)	(5)
EAWA $\times$ STEM $\times$ TARP	0.073** (2.16)	0.081** (2.29)	0.072** (2.29)	0.049* (1.87)	0.056** (2.05)
EAWA $\times$ TARP	-0.030** (-2.21)	-0.044 (-1.52)	-0.034** (-2.40)	-0.020* (-1.92)	-0.035 (-1.32)
Post $\times$ STEM $\times$ TARP	0.016 (0.60)	0.022 (0.68)	0.008 (0.32)	0.005 (0.23)	0.012 (0.42)
Post $\times$ TARP	0.005 (0.24)	-0.065** (-2.20)	0.013 (0.55)	0.021 (1.11)	-0.030 (-0.90)
TARP	-0.058 (-1.43)	-0.014 (-0.45)			
STEM	-0.008 (-0.53)	-0.006 (-0.35)	0.004 (0.79)	0.006 (0.98)	0.010 (1.56)
STEM $\times$ TARP	-0.011 (-0.58)	-0.013 (-0.70)	-0.020* (-1.89)	-0.007 (-0.62)	-0.010 (-1.08)
Constant	1.321*** (36.83)	1.318*** (35.61)	1.265*** (107.51)	1.254*** (109.30)	1.280*** (59.41)
Bank FE	No	No	Yes	Yes	Yes
City FE	No	No	No	Yes	Yes
Year $\times$ Month FE	No	Yes	No	No	Yes
N	8,981	8,981	8,981	8,981	8,981
adj. R ²	0.01	0.02	0.08	0.19	0.20

Online Appendix: Nationalistic Labor Policies Hinder Financial Innovation

Francesco D’Acunto, Hengyi Huang, Michael Weber, Jin Xie, and Liu Yang

Not for Publication

A.1 Institutional Setting

Our empirical setting is influenced by two key institutional features: the timing and motivations governing the approval of TARP and EAWA, and the scope for patenting activity within the US financial sector.

A.1.1 TARP and Employ American Workers Act

On October 3, 2008, US President George W. Bush enacted the Troubled Asset Relief Program (TARP), granting the US government the authority to acquire toxic assets and equity from financial institutions.¹ Predominantly consisting of commercial banks, TARP participants saw the government acquiring preferred stocks in eight major banks: Goldman Sachs, Morgan Stanley, J.P. Morgan Chase, Bank of America, Citigroup, Wells Fargo, Bank of New York Mellon, and State Street. TARP assistance also extended to a major utility (General Electric) and three leading car manufacturers (GM, Ford, and Chrysler). We exclude these two groups from our analysis, due to the distinct nature of their innovation and their extensive coverage by various government programs during the financial crisis and the Great Recession.

On February 17, 2009, President Obama signed the Employ American Workers Act (EAWA) into law, which was valid for two years and was not subsequently renewed. The legislation specifically targeted TARP recipients seeking to hire H-1B workers. Although the original intent of the Act was to completely prohibit TARP recipients from applying for any H-1B visas, the final legislation permitted such employment with certain imposed restrictions. Irrespective of their exposure and reliance on H-1B employees, all TARP recipients were subjected to the same regulations as “H-1B Dependent Employers.”² The foreign hiring ban stipulated by the EAWA did not extend to workers seeking to extend their H-1B stay; however, it was binding for employers filing H-1B petitions for new foreign-born employees.

In the US, an employer with a significant proportion of H-1B workers relative to its total domestic workforce is categorized as “H-1B dependent.” Such employers are required to make specific attestations on the Labor Condition Application (LCA). However, under the EAWA, TARP recipients are also treated

¹Originally, the program authorized an expenditure of US\$700 billion. The Dodd-Frank Wall Street Reform and Consumer Protection Act later reduced this amount to US\$475 billion. The program essentially concluded on December 19, 2014, when the US Treasury sold its remaining holdings of Ally Financial.

²Certain non-TARP recipients are considered “H-1B dependent” if H-1B workers make up more than 15% of their workforce.

as H-1B dependent employers in relation to *new* petitions filed, irrespective of the number of H-1B workers in their US workforce. Precisely, the EAWA stipulates that no TARP recipient can hire an H-1B non-immigrant unless the recipient has made certain LCA attestations, which were previously mandated only for H-1B dependent employers.

The extra obligations required by EAWA include the following:

- Prior to filing an H-1B petition, the employer must take *bona fide* steps to recruit US workers for the position for which an H-1B worker is sought and offer a wage that is at least as high as that offered to the H-1B worker.
- The employer must also attest that, in connection with their *bona fide* recruitment efforts, they have actually offered the same job to any US worker who has applied and who is equally or better qualified for the position.
- Employers were also hindered from laying off any US workers in jobs equivalent to the H-1B position within the period beginning 90 days before and ending 90 days after the H-1B petition filing.

In summary, H-1B petitioning under EAWA imposed significantly higher compliance costs, requiring detailed information and documents for each prospective worker. Additionally, conducting a documented search for US applicants with specific skills, if rare within the domestic workforce, could be both costly and time-consuming. As demonstrated below, the EAWA ban indeed resulted in the near-complete elimination of new foreign high-skilled hires by US financial institutions.

A.1.2 Patenting in the Financial Sector

Our primary focus is on the innovation of financial institutions. Although the innovation activities of traditional banks are sometimes overlooked, note that banks are frequent patent filers (e.g., [Chen et al., 2019](#); [Lerner et al., 2023](#)). The aftermath of the 2008-2009 financial crisis witnessed an unprecedented surge in patenting activity by commercial banks in the US. According to a *Wall Street Journal* article dated May 10, 2016, large banks and credit companies have submitted applications for at least 2,679 patents since 2013, particularly in areas such as mobile systems, the Internet of Things, and data analytics.³

To illustrate the areas in which financial institutions, particularly banks, have been innovative, we categorize granted patents based on US classifications and rank these classifications by the number of patents within each category. In Panel A of Table 1, we present the top 20 classifications of patents filed by the financial institutions in our sample, utilizing the US Patent and Trademark Office (USPTO) patent

³“Big Banks Stake Fintech Claims with Patent Application Surge,” *The Wall Street Journal*, May 10, 2016.

topic classifications. These classifications account for approximately 95% of patents filed by commercial banks headquartered in the US. Common categories include various forms of data processing methods and information security.⁴

Financial institutions are driven by incentives to participate in FinTech innovation for several reasons. First, due to security threats, the risk of data breaches, and potential litigation, banks cannot rely solely on third-party vendors to develop patents. Second, FinTech companies pose competition to incumbent traditional financial institutions in various areas, including mortgage and loan originations. Banks have actively filed patents over the past decade in domains where startup FinTech firms are flourishing. Some of these inventions have already been commercialized into financial products. For instance, the cloud wallet security technology developed by J.P. Morgan Chase has been integrated into the Chase Pay mobile payments system. On the other hand, some patent applications by financial institutions are more speculative. For example, MasterCard International has applied for a patent on a method that utilizes big data to predict customers’ political affiliations based on their shopping patterns.⁵

To provide additional insight into the type of assignees and inventions covered by the patents in our sample, Panel B of Table 1 presents information on the assignee and abstract for the top 20 patents by citations in our dataset. For instance, the most cited patent, filed by J.P. Morgan Chase in March 2012, introduces a “method and system for processing internet payments using the electronic funds transfer network.” Additionally, a series of patents filed by American Express, focusing on customer loyalty reward systems, are among the most cited. In summary, this table highlights patents that broadly address payment systems, the cybersecurity of financial institutions’ client data, and online customer loyalty programs.

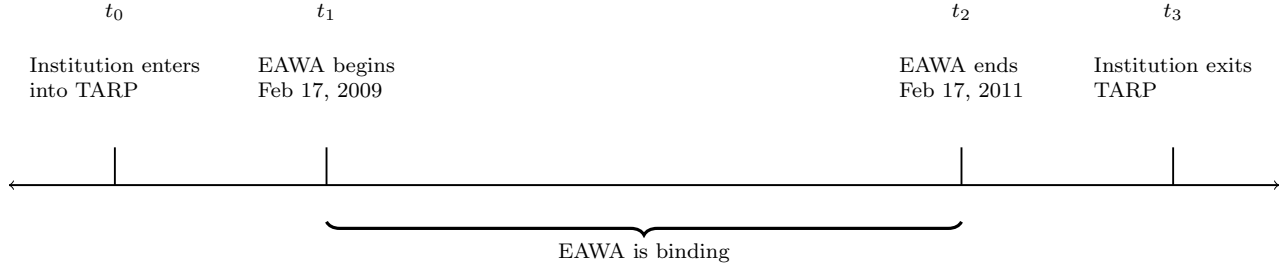
Figure A.4 in the Online Appendix presents the front page of the patent with the highest citation among all patents in our sample. This business-method patent, granted to JPMorgan Chase Bank, introduces a method and system for conducting financial transactions over a payment network. Figure A.5 in the Online Appendix showcases the front page of the patent with the highest citation among all FinTech patents in our sample. This particular patent, granted to Bank of America, focuses on inventing hardware for attesting to authenticity in a cloud environment.

⁴The catch-all category includes patents that we cannot unequivocally assign to one of the USPTO categories.

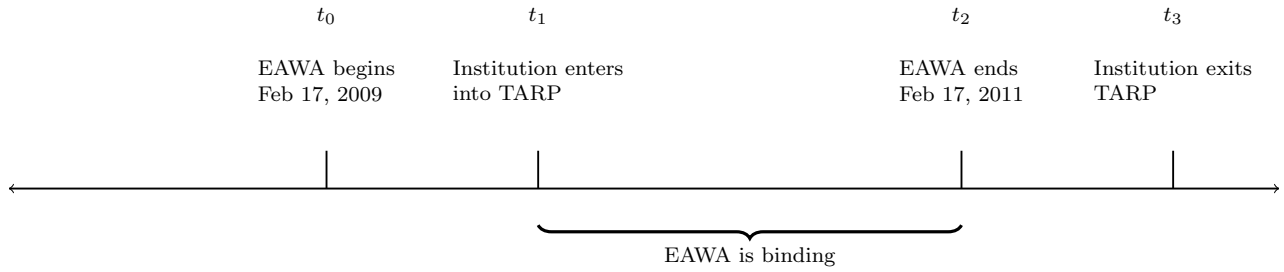
⁵See “Big Banks Stake FinTech Claims with Patent Application Surge,” *Wall Street Journal*, May 10, 2016.

Figure A.1: Timelines

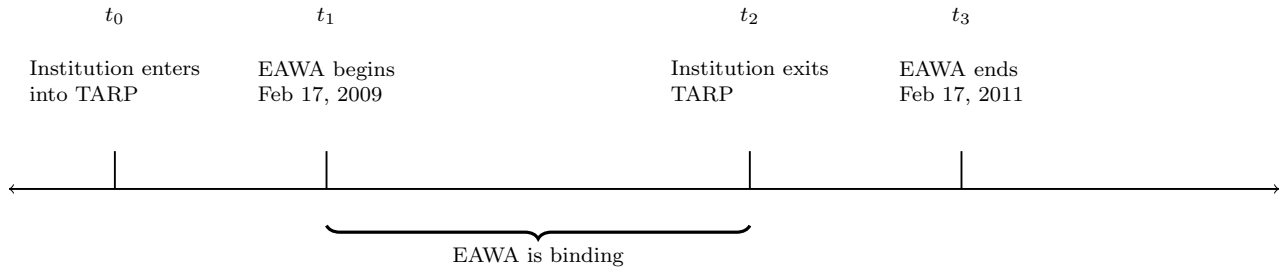
Panel A: Financial institution enters into TARP before Feb 17, 2009 & exits TARP after Feb 17, 2011



Panel B: Financial institution enters into TARP after Feb 17, 2009 & exits TARP after Feb 17, 2011



Panel C: Financial institution enters into TARP before Feb 17, 2009 & exits TARP before Feb 17, 2011



Panel D: Financial institution enters into TARP after Feb 17, 2009 & exits TARP before Feb 17, 2011

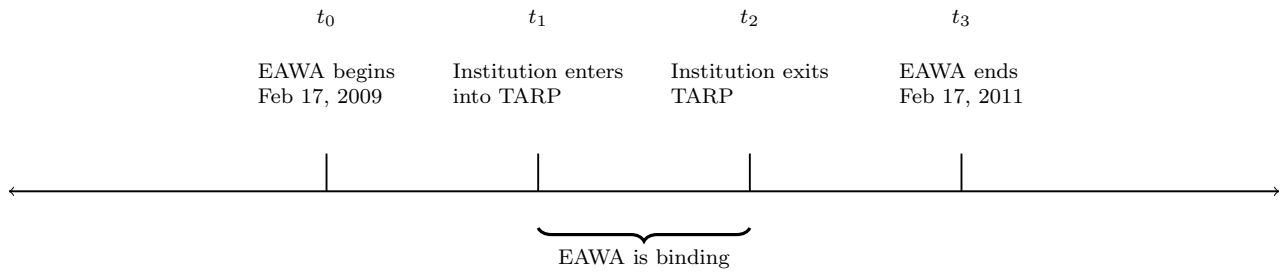


Figure A.2: **Top 20 Financial Institutions Ranked by FinTech Patents**

This figure presents the name of financial institutions (and their corresponding numbers) that filed the most patents from January 2007 through June 2015. These patents were granted by the US Patent and Trademark Office. The x-axes are institution names. The y-axes are the number of FinTech patents filed by institutions.

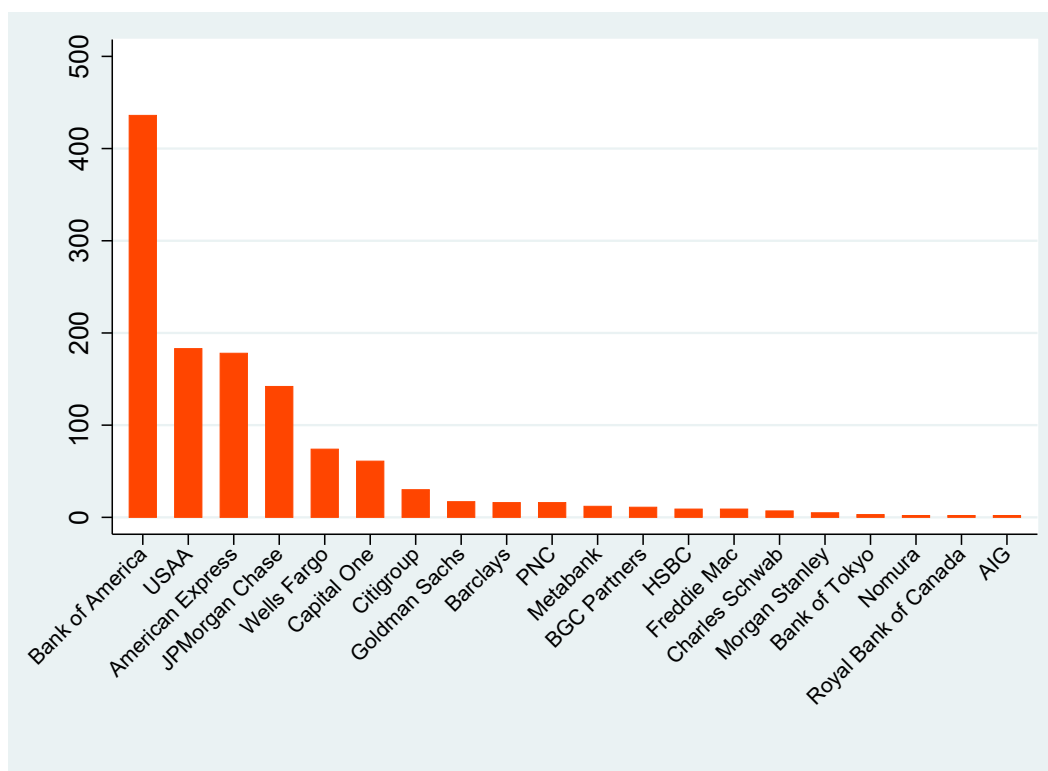


Figure A.3: **Distribution of Patent Citations**

This figure plots the distribution of patent citations across patents filed by sample financial institutions from January 2007 through December 2014. These patents were granted by the US Patent and Trademark Office. Panel A plots the distribution of the number of citation counts. Panel B plots the distribution of adjusted citations. Adjusted citation is the number of citations divided by the average number of citations in a given technology-class-year (the year in which all patents were applied).

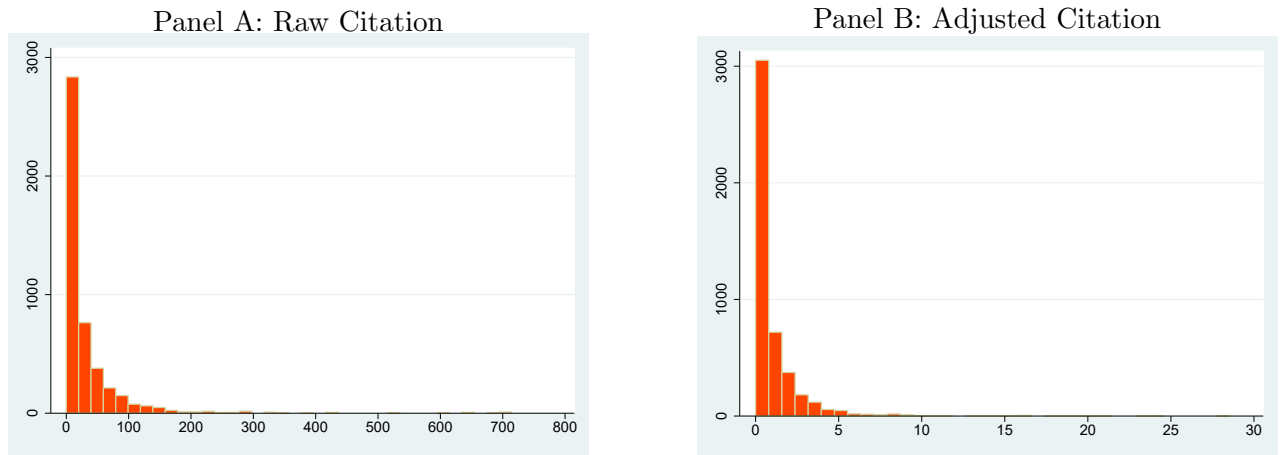


Figure A.4: Front Page of Patent with the Highest Citation Count



US008738521B2

(12) United States Patent O'Leary et al.	(10) Patent No.: US 8,738,521 B2 (45) Date of Patent: *May 27, 2014
---	--

(54) METHOD AND SYSTEM FOR PROCESSING INTERNET PAYMENTS USING THE ELECTRONIC FUNDS TRANSFER NETWORK (71) Applicant: JPMorgan Chase Bank, N.A., New York, NY (US) (72) Inventors: Denis O'Leary, New York, NY (US); Vincent D'Agostino, New York, NY (US); S. Richard Re, Westfield, NJ (US); Jessica Burney, New York, NY (US); Adam Hoffman, New York, NY (US) (73) Assignee: JPMorgan Chase Bank, N.A., New York, NY (US) (*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days. This patent is subject to a terminal disclaimer. (21) Appl. No.: 13/789,795 (22) Filed: Mar. 8, 2013 (65) Prior Publication Data US 2013/0191277 A1 Jul. 25, 2013 Related U.S. Application Data (63) Continuation of application No. 13/443,179, filed on Apr. 10, 2012, now Pat. No. 8,452,703, which is a continuation of application No. 13/102,113, filed on May 6, 2011, now Pat. No. 8,190,521, which is a continuation of application No. 12/576,463, filed on Oct. 9, 2009, now Pat. No. 7,962,409, which is a continuation of application No. 10/356,171, filed on Jan. 31, 2003, now Pat. No. 7,676,431, which is a continuation of application No. 09/497,307, filed on Feb. 3, 2000, now Pat. No. 6,609,113. (60) Provisional application No. 60/132,305, filed on May 3, 1999. (51) Int. Cl. G06Q 20/10 (2012.01) G06Q 40/00 (2012.01) (52) U.S. Cl. USPC 705/39; 705/35; 705/42	(58) Field of Classification Search CPC G06Q 20/10; G06Q 40/00 USPC 705/39, 35 See application file for complete search history. (56) References Cited U.S. PATENT DOCUMENTS <table border="0" style="width: 100%;"> <tr> <td style="width: 30%;">3,852,571 A</td> <td style="width: 30%;">12/1974</td> <td style="width: 40%;">Hall et al.</td> </tr> <tr> <td>4,321,672 A</td> <td>3/1982</td> <td>Braun et al.</td> </tr> </table> (Continued) FOREIGN PATENT DOCUMENTS <table border="0" style="width: 100%;"> <tr> <td style="width: 30%;">EP</td> <td style="width: 30%;">04/21808</td> <td style="width: 40%;">5/1990</td> </tr> <tr> <td>WO</td> <td>97/45814</td> <td>4/1997</td> </tr> </table> (Continued) OTHER PUBLICATIONS Penny, Jonathan; The Payment Revolution: The Growth of person-to-person Generation Y Payments services; Journal of Financial Services Marketing; Dec. 2001; 6,2; p. 190.* (Continued) <i>Primary Examiner</i> — Jagdish Patel (74) <i>Attorney, Agent, or Firm</i> — Goodwin Procter LLP (57) ABSTRACT Embodiments of the invention include a method and system for conducting financial transactions over a payment network. The method may include associating a payment address of an account with an account holder name, the account residing at a financial institution and the associated payment address of the account configured to allow withdrawals by the account holder only and to allow a plurality of deposits to be made at different times. The method further includes freely publishing the payment address and making it available to users of an internet portal or search engine. The method further includes receiving data over a network identifying a deposit to be made to the account, assigning the deposit to the account using the payment address, and notifying the payer of the assignment. At least one directory is used for associating the account holder with the payment address. 24 Claims, 10 Drawing Sheets	3,852,571 A	12/1974	Hall et al.	4,321,672 A	3/1982	Braun et al.	EP	04/21808	5/1990	WO	97/45814	4/1997
3,852,571 A	12/1974	Hall et al.											
4,321,672 A	3/1982	Braun et al.											
EP	04/21808	5/1990											
WO	97/45814	4/1997											

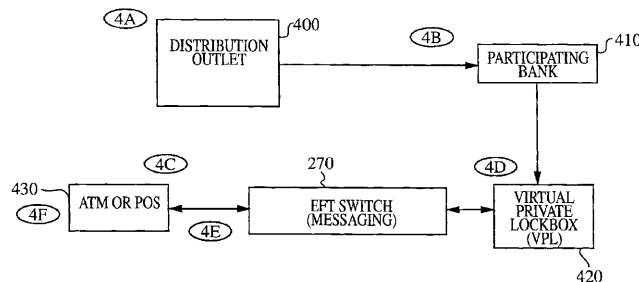


Figure A.5: **Front Page of FinTech Patent with the Highest Citation Count**



US009184918B2

(12) **United States Patent**
Spiers et al.

(10) **Patent No.:** **US 9,184,918 B2**

(45) **Date of Patent:** *Nov. 10, 2015

(54) **TRUSTED HARDWARE FOR ATTESTING TO AUTHENTICITY IN A CLOUD ENVIRONMENT**

(58) **Field of Classification Search**
None
See application file for complete search history.

(71) Applicant: **Bank of America Corporation,**
Charlotte, NC (US)

(56) **References Cited**

(72) Inventors: **Bradford Thomas Spiers**, Bedminster, NJ (US); **Miroslav Halas**, Charlottesville, VA (US); **Richard A. Schimmel**, Glenmont, NY (US)

U.S. PATENT DOCUMENTS

7,146,305	B2	12/2006	van der Made
7,207,039	B2	4/2007	Komarla et al.

(Continued)

(73) Assignee: **Bank of America Corporation,**
Charlotte, NC (US)

FOREIGN PATENT DOCUMENTS

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

CN	102033755	A	4/2011
EP	1761837	A1	3/2007
JP	2007081014	A	3/2007
WO	2006011943	A1	2/2006

This patent is subject to a terminal disclaimer.

OTHER PUBLICATIONS

(21) Appl. No.: 14/305,611

Siqin Zhao, et al. "The Application of Virtual Machines on System Security," 2009 Fourth ChinaGrid Annual Conference, DOI 10.1109/ChinaGrid.2009.45, Sep. 2009, pp. 222-229.

(22) Filed: **Jun. 16, 2014**

(Continued)

(65) **Prior Publication Data**

US 2014/0298439 A1 Oct. 2, 2014

Related U.S. Application Data

Primary Examiner — Fatoumata Traore

(74) *Attorney, Agent, or Firm* — Banner & Witcoff, Ltd.;
Michael A. Springs

(63) Continuation of application No. 13/422,751, filed on Mar. 16, 2012, now Pat. No. 8,839,363.

(60) Provisional application No. 61/492,612, filed on Jun. 2, 2011, provisional application No. 61/476,747, filed on Apr. 18, 2011.

(57) **ABSTRACT**

(51) **Int. Cl.**
H04L 29/06 (2006.01)
H04L 9/32 (2006.01)
 (Continued)

Apparatuses, computer readable media, methods, and systems are described for storing a first measurement of a virtualization platform, storing a second measurement of a measured virtual machine, generating a quote using a key, wherein the quote is based on the first measurement and the second measurement, and providing the quote for attesting to authenticity of the virtualization platform and of the measured virtual machine. In a further example, the quote may be generated based on a third measurement of a secure tunnel.

(52) U.S. Cl.
CPC *H04L 9/3247* (2013.01); *G06F 9/45533*
(2013.01); *G06F 21/575* (2013.01); *H04L 9/28*
(2013.01); *H04L 9/3234* (2013.01);

(Continued)

20 Claims, 9 Drawing Sheets

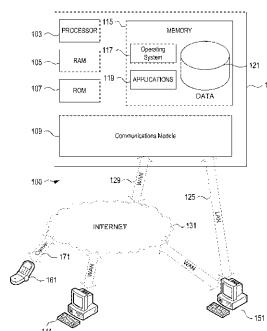


Table A.1: TARP Banks in Our Sample

This table presents a list of 27 TARP banks that filed at least one patent before December 31, 2006. #H-1B STEM 2004-2006 refers to the number of H-1B STEM workers that a bank had hired during 2004-2006. TARP Start refers to the date on which a financial institution agrees to receive funds from the Treasury. TARP End refers to the date on which a financial institution fully pays off the funds. EAWA Start and EAWA End refer to dates on which the EAWA act starts and ends to be ineffective for a bank. #EAWA Days is the number of days for which a TARP participant has been subject to EAWA. Assets 2009 (in millions) is a bank's total assets as of 2009.

	#H-1B STEM 2004-2006	TARP Start	TARP End	EAWA Start	EAWA End	#EAWA Days	Assets 2009
JPMorgan Chase	848	2008/10/28	2009/12/16	2009/02/17	2009/12/16	302	2,175,052
Chase Manhattan Bank	0	2008/10/28	2009/12/16	2009/02/17	2009/12/16	302	2,031,989
Citigroup	1,629	2008/10/28	2011/01/31	2009/02/17	2011/01/31	713	1,938,470
Bank of America	416	2008/10/28	2010/03/09	2009/02/17	2010/03/09	385	1,817,943
Wells Fargo	461	2008/10/28	2010/05/26	2009/02/17	2010/05/26	463	1,243,646
Morgan Stanley	718	2008/10/28	2009/08/12	2009/02/17	2009/08/12	176	658,812
Wachovia	245	2008/10/28	2010/05/26	2009/02/17	2011/02/17	730	510,083
Washington Mutual	134	2008/10/28	2009/12/16	2009/02/17	2011/02/17	730	327,913
PNC Financial Services Group	0	2008/12/31	2010/05/05	2009/02/17	2010/05/05	442	291,081
US Bancorp	44	2008/11/14	2009/07/15	2009/02/17	2009/07/15	148	265,912
Merrill Lynch Inc	301	2008/10/28	2010/03/09	2009/02/17	2010/03/09	385	246,024
Bank of New York Mellon	0	2008/10/28	2009/08/05	2009/02/17	2009/08/05	169	237,512
Capital One	14	2008/11/14	2009/12/09	2009/02/17	2009/12/09	295	169,646
Branch Banking and Trust Company	5	2008/11/14	2009/07/22	2009/02/17	2009/07/22	155	147,484
American Express	100	2009/01/09	2009/07/29	2009/02/17	2009/07/29	162	126,074
KeyCorp	14	2008/11/14	2011/04/20	2009/02/17	2011/02/17	730	105,231
Goldman Sachs	1,266	2008/10/28	2009/07/22	2009/02/17	2009/07/22	155	91,016
Zions Bancorporation	0	2008/11/14	2012/12/05	2009/02/17	2011/02/17	730	55,093
Huntington Bancshares	0	2008/11/14	2011/01/19	2009/02/17	2011/01/19	701	54,353
Fifth Third Bank	3	2008/12/31	2011/03/16	2009/02/17	2011/02/17	730	54,116
Discover Financial Services	90	2009/03/13	2010/07/07	2009/03/13	2010/07/07	481	39,892
TCF Financial Corporation	5	2008/11/14	2009/12/21	2009/02/17	2009/12/21	307	16,740
Citizens Bank	11	2009/05/29	2013/03/26	2009/05/29	2011/02/17	629	12,267
Silicon Valley Bank	18	2008/12/12	2010/06/16	2009/02/17	2010/06/16	484	9,474
First American Financial Corporation	0	2009/07/24	2012/12/11	2009/07/24	2011/02/17	573	5,530
First American Corporation	0	2009/07/24	2012/12/11	2009/07/24	2011/02/17	573	218
Bank One	57	2008/10/28	2009/12/16	2009/02/17	2009/12/16	302	215

Table A.2: **FinTech Keywords**

This table presents the full keyword list to identify the following seven types of FinTech patents. We search these keywords in the abstracts and claims for each patent.

Fintech Class	Keywords	Fintech Class	Keywords
Cybersecurity	cybersecurity	Blockchain	blockchain
Cybersecurity	encryption	Blockchain	distributed ledger
Cybersecurity	tokenization	Blockchain	cryptocurrency
Cybersecurity	authentication	Blockchain	acyclic
Cybersecurity	biometrics	Blockchain	bitcoin
Moblie Transaction	mobile transaction	Peer-to-peer	peer-to-peer
Moblie Transaction	payment	Peer-to-peer	peer
Moblie Transaction	digital wallet	Peer-to-peer	p2p
Moblie Transaction	digital cash	Peer-to-peer	consumer-to-consumer
Moblie Transaction	virtual cash	Peer-to-peer	customer-to-customer
Moblie Transaction	automated clearing house	Peer-to-peer	crowdfunding
Moblie Transaction	automatic funds transfer	Peer-to-peer	crowd funding
Moblie Transaction	automatic investment program	Robo-advising	robo-advising
Moblie Transaction	automatic reinvestment	Robo-advising	automatic
Moblie Transaction	electronic depository transfers	Robo-advising	portfolio
Moblie Transaction	electronic funds transfer	Robo-advising	future investment opportunities
Data Analytics	data analytics	Robo-advising	investment adviser
Data Analytics	big data	Robo-advising	investment advisory
Data Analytics	cloud computing	Robo-advising	investment strategy
Data Analytics	artificial	Robo-advising	market timing
Data Analytics	machine learning	Robo-advising	passive investment strate
Data Analytics	credit history	Robo-advising	passive portfolio strate
Data Analytics	credit scoring	Robo-advising	replicating portfolio
Blockchain	crypto currency	Robo-advising	well-diversified portfolio
Blockchain	digital currency	Internet of things	internet of things
Blockchain	digital currencies	Internet of things	smart devise
Blockchain	virtual currency	Internet of things	sensor
Blockchain	virtual currencies	Internet of things	actuators

Table A.3: Patent Citations Data: Descriptive Statistics

This table presents descriptive statistics for our sample of patent citations. The sample unit is bank i as of year-month s in which at least one granted patent was filed. The sample period is 2007-2014. We require banks to have submitted at least one patent application with the US Patent and Trademark Office (USPTO) before December 31, 2006. Panel A includes 20 TARP and 17 non-TARP banks, both of which had hired at least one H-1B STEM worker during 2004-2006. Panel B includes 27 TARP banks. TARP is a dummy variable that equals 1 if bank i had ever accessed the Troubled Asset Relief Program (TARP), and 0 otherwise. EAWA is a dummy variable that equals 1 if bank i complied with the Employ American Workers Act (EAWA) in year-month s , and 0 otherwise. Post is a dummy variable that equals 1 for all year-months after the EAWA was ineffective, and 0 otherwise. For each patent category (i.e., all patents, business method (705), non-business method (Non-705), and FinTech), $\overline{\text{Raw Cites}}$ is the averaged raw citation counts for all patents filed by bank i in year-month s that were later granted. $\overline{\text{Adj Cites}}$ is the averaged adjusted citation counts for all patents filed by bank i in year-month s that were later granted. For each patent, adjusted citation is the number of citations divided by the average number of citations in a given technology-class-year (the year in which all patents of the same technology were applied). Technology domain is defined at the 4-digit International Patent Classification. The end of the observation period for patent citation is January, 2022. $H1B_{-3}$ is a dummy variable that equals 1 if bank i as of year-month s hired at least one H-1B-sponsored worker over the last three years, and 0 otherwise. $STEM_{-3}$ is a dummy variable that equals 1 if bank i as of year-month s hired at least one H-1B-sponsored STEM worker over the last three years, and 0 otherwise. TARP Fund% is the amount of funding over total assets measured in year 2009 that TARP bank i obtained from the government through entering into the TARP program. High Exposure_i is a dummy variable that equals 1 if the number of H-1B STEM workers hired by bank i during 2004-2006 is above the median of the distribution over a subsample of banks of similar size. Banks are classified into small, medium, and large if their assets are below US\$10 billion, between US\$10 and US\$50 billion, and above US\$1 trillion, respectively.

	N	Mean	Std	Min	P10	P25	P50	P75	P90	Max
Panel A. All Banks										
TARP	651	0.897	0.304	0	0	1	1	1	1	1
EAWA	651	0.088	0.283	0	0	0	0	0	0	1
Post	651	0.556	0.497	0	0	0	1	1	1	1
$\overline{\text{Raw Cites}}$	651	38.877	51.373	0	3	9	22.333	49	94	643
$\overline{\text{Raw Cites}}$ (705)	404	50.128	70.812	0	3.750	9	24	58.917	127	643
$\overline{\text{Raw Cites}}$ (Non-705)	493	36.202	43.832	0	3	9.667	21.500	45.333	93.600	391
$\overline{\text{Raw Cites}}$ (FinTech)	354	43.334	59.495	0	3	9.333	22.111	53	107	520
$\overline{\text{Adj Cites}}$	651	1.034	1.258	0	0.089	0.255	0.636	1.319	2.411	11.650
$\overline{\text{Adj Cites}}$ (705)	404	1.114	1.548	0	0.081	0.193	0.510	1.370	2.803	11.650
$\overline{\text{Adj Cites}}$ (Non-705)	493	1.101	1.259	0	0.095	0.321	0.680	1.341	2.776	8.998
$\overline{\text{Adj Cites}}$ (FinTech)	354	1.307	1.727	0	0.084	0.308	0.699	1.603	3.461	15.460
$H1B_{-3}$	651	0.980	0.140	0	1	1	1	1	1	1
$STEM_{-3}$	651	0.975	0.155	0	1	1	1	1		
Panel B. TARP Banks										
EAWA	641	0.105	0.306	0	0	0	0	0	1	1
Post	641	0.618	0.486	0	0	0	1	1	1	1
$\overline{\text{Raw Cites}}$	641	38.736	50.978	0	4	9.500	22.333	49	93.778	643
$\overline{\text{Raw Cites}}$ (705)	404	51.207	70.522	0	4	9.875	25.917	60.250	127	643
$\overline{\text{Raw Cites}}$ (Non-705)	484	35.057	43.048	0	3	9	20.750	44	87.500	391
$\overline{\text{Raw Cites}}$ (FinTech)	348	41.877	59.109	0	2.778	9	21.625	52	102.667	520
$\overline{\text{Adj Cites}}$	641	1.027	1.242	0	0.100	0.258	0.636	1.308	2.355	11.654
$\overline{\text{Adj Cites}}$ (705)	404	1.141	1.545	0	0.089	0.208	0.546	1.424	2.805	11.654
$\overline{\text{Adj Cites}}$ (Non 705)	484	1.069	1.238	0	0.097	0.313	0.651	1.298	2.729	8.998
$\overline{\text{Adj Cites}}$ (FinTech)	348	1.269	1.724	0	0.076	0.287	0.677	1.555	3.359	15.459
$H1B_{-3}$	641	0.910	0.287	0	1	1	1	1	1	1
$STEM_{-3}$	641	0.899	0.302	0	0	1	1	1	1	1
TARP Fund%	641	0.027	0.028	0.009	0.011	0.014	0.020	0.026	0.063	0.110
High Exposure	641	0.891	0.312	0	0	1	1	1	1	1

Table A.4: **Employ American Workers Act (EAWA) and First Patent Filers as Lead Inventors**

This table reports the estimates of the effect of EAWA on the likelihood of patents being filed by first-time filers. Columns (1)-(4) report estimates from the following ordinary least squares (OLS) specification:

$$\text{First Filer Lead}_{i,s} = \alpha + \beta_1 \text{EAWA}_{i,s} + \beta_2 \text{EAWA}_{i,s} \times \text{TARP}_i + \beta_3 \text{Post}_{i,s} + \beta_4 \text{Post}_{i,s} \times \text{TARP}_i + X'_i \times \theta + \eta_i + \eta_s + \epsilon_{i,s}.$$

The sample in columns (1)-(4) consists of 20 TARP and 13 non-TARP banks that sponsored at least one STEM job through the H-1B program during 2004-2006. Columns (5)-(8) report estimates from the following OLS specification:

$$\text{First Filer Lead}_{i,s} = \alpha + \beta_1 \text{EAWA}_{i,s} + \beta_2 \text{EAWA}_{i,s} \times \text{High Exposure}_i + \beta_3 \text{Post}_{i,s} + \beta_4 \text{Post}_{i,s} \times \text{High Exposure}_i + X'_i \times \theta + \eta_i + \eta_s + \epsilon_{i,s}.$$

The sample in columns (5)-(8) consists of 27 TARP banks. The dependent variable is First Filer Lead_{i,s}, which is a dummy variable that equals 1 if patents filed by bank *i* in year-month *s* include at least one inventor who not only filed for the first time for bank *i* but also was listed as the lead inventor, and 0 otherwise. See Table 2 for definitions of other variables. Standard errors are clustered at the bank level.

	All Banks				TARP Banks			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
EAWA × TARP	-0.075*** (-2.90)	-0.085** (-2.50)	-0.059** (-2.44)	-0.047 (-1.69)				
Post × TARP	-0.014 (-0.56)	-0.018 (-0.58)	-0.054*** (-2.74)	-0.033 (-1.33)				
EAWA × High Exposure					-0.068* (-1.83)	-0.070* (-1.88)	-0.038 (-1.20)	-0.039 (-1.27)
EAWA					-0.043 (-1.27)	-0.025 (-0.48)	-0.050 (-1.69)	-0.056 (-1.39)
Post × High Exposure					-0.028 (-0.78)	-0.026 (-0.73)	-0.042* (-1.82)	-0.042* (-1.80)
Post					0.007 (0.20)	0.035 (0.51)	-0.016 (-0.91)	-0.012 (-0.34)
EAWA × TARP Fund%					1.200 (1.53)	1.226 (1.57)	1.258 (1.62)	1.276 (1.55)
Post × TARP Fund%					-0.037 (-0.07)	-0.037 (-0.06)	-0.242 (-1.12)	-0.263 (-1.21)
TARP	0.126** (2.33)	0.130** (2.11)						
High Exposure					0.086 (1.68)	0.083 (1.60)		
TARP Fund%					-1.221 (-1.25)	-1.229 (-1.21)		
H1B ₋₃	0.049 (1.60)	0.059* (1.70)	-0.006 (-0.68)	0.004 (0.28)	0.052 (1.66)	0.049 (1.55)	0.020* (2.02)	0.022* (1.79)
STEM ₋₃	0.060** (2.30)	0.056** (2.04)	-0.014* (-2.03)	-0.028*** (-2.87)	0.052 (1.06)	0.058 (1.05)	-0.053*** (-5.00)	-0.058*** (-3.65)
Constant	-0.065* (-1.76)	-0.071* (-1.73)	0.125*** (8.50)	0.119*** (6.69)	0.025 (0.68)	0.006 (0.11)	0.158*** (9.75)	0.159*** (7.11)
Year × Month FE	No	Yes	No	Yes	No	Yes	No	Yes
Bank FE	No	No	Yes	Yes	No	No	Yes	Yes
N	3,168	3,168	3,168	3,168	2,592	2,592	2,592	2,592
adj. R ²	0.06	0.05	0.43	0.43	0.06	0.04	0.42	0.42